Humboldt-Universität zu Berlin

Masterarbeit

Characterization of cylindrical wavelength shifting optical light guides.
Charakterisierung von zylindrischen wellenlängenschiebenden optischen Lichtleitern.

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The wavelength shifting optical module (WOM) is a new single photon detector for the UV regime with high signal-to-noise ratio specified for the IceCube neutrino telescope. Thirty-five cylindrical light guides have been coated with a wavelength shifting paint to investigate the influence of the geometry on the efficiency. A semi-automatized calibration setup was developed and twenty-two of the coated light guides were characterized. No dependencies of the diameter or wall thickness on the efficiency of the light guide were found and no difference between light guides produced with polymethylmethacrylat (PMMA) and quartz glass was observed. It was found that the light distribution on the ends of the PMMA light guide can be seen as homogeneous if the illumination point has a distance of ≥ 50 mm. The detection efficiencies of the different WOMs varied with a standard deviation of $\pm 0.8\,\%$ around the average of $(8.49\ \pm 0.01_{\mathrm{stat}}\ ^{+0.41}_{-0.55\,\mathrm{sys}})$ %. It was concluded that the coated paint thickness does not provide for the measured WOMs full UV absorption.

Charakterisierung von zylindrischen wellenlängenschiebenden optischen Lichtleitern.

Das wellenlängenschiebende optische Modul (WOM) ist ein neuer Einzelphotonendetektor für den UV-Bereich mit hohem Signal-Rausch-Verhältnis entwickelt für das Neutrinoteleskop IceCube. Fünfunddreißig zylindrische optische Lichtleiter wurden mit wellenlängenschiebender Farbe beschichtet um den Einfluss der Geometrie auf die Sammeleffizienz zu untersuchen. Ein teilautomatisierter Kalibrationsaufbau wurde entwickelt, mit welchem zweiundzwanzig dieser Lichtleiter charakterisiert wurden. Es wurden keine Abhängikeiten der Effizienz von dem Durchmesser oder der Wandstärke des Leiters gefunden. Weiter wurden keine Unterschiede bei der Verwendung von Polymethylmethacrylat (PMMA) and Quartzglass als Material für den Lichtleiter beobachtet. Es wurde herausgefunden, dass die Lichverteilung an den Schnittkanten im Falle des PMMA als homogen betrachtet werden kann, wenn der Punkt der Beleuchtung ≥ 50 mm entfernt ist. Die Sammeleffizienz der Lichtleiter variierte mit einer Standardabweichung von $\pm 0.8\,\%$ um den Mittelwert von $(8.49\ \pm 0.01_{\mathrm{stat}}\ ^{+0.41}_{-0.55\,\mathrm{sys}})$ %. Es wurde geschlussfolgert, dass die aufgetragene Farbdicke nicht ausreicht um bei den gemessenen Lichtleitern volle UV-Absorption zu erreichen.

Inhaltsverzeichnis

1 Introduction **1**

2 Theory **3**

2.1 The Wavelength Shifting Optical Module (WOM) 3

2.1.1 Theoretical capture efficiency 4

2.2 Photomultiplier tube 7

2.2.1 Counting photons - The model-independent approach 9

3 Coating **11**

3.1 Wavelength shifting paint 12

3.2 Coating speed 13

3.3 Coating process 14

3.4 Paint thickness 15

3.5 Seals 17

4 Efficiency measurement setup **19**

4.1 Five Axis Scanner 20

4.1.1 Control algorithm 21

4.2 Light source 22

4.2.1 Opening angle 23

4.2.2 Stability 24

4.3 Functional demonstration with an IceCube-PMT 25

5 WOM Characterizations **27**

5.1 Raw data analysis 27

5.1.1 Waveform integration 27

5.1.2 Offset correction 28

5.2 Measurement of the WOMs' efficiency 30

5.2.1 Measurement procedure 30

5.2.2 Correction factor to involve the spectral sensitivity 30

5.2.3 Determination of the position-dependent capture efficiencies 31

5.2.4 Radial symmetry 32

5.3 Geometry 36

5.4 Systematic uncertainties 38

5.4.1 Photocathodes' spatial variation 38

5.4.2 PMT linearity 42

5.4.3 Nonlinear shift of the light source 44

5.5 Estimation of the overall capture efficiency 45

6 Cross check with different methods **49**

6.1 Setup Mainz 49

6.2 photocathode's spatial variation setup Mainz 49

6.3 Results 51

7 Summary **53**

Appendices **57**

A Appendix **57**

A.1 Results of all investigated WOMs 59

A.2 Instructions to use WACKER SilGel® 612 A/B for optical contact . . . 60

A.3 Derivation of the torque from tube contraction 62

A.4 Technical drawings 64

Bibliography **67**

1. Introduction

A new detector device is developed.

IceCube is a neutrino telescope to detect interstellar neutrinos in the range of GeV to PeV. The detector measures the Cherenkov light produced by charged particles in the South Pole ice originated by the neutrinos' weak interaction with the ice. The Cherenkov light is detected with digital optical modules (DOM) buried in ice. A DOM consists of a 10-inch photomultiplier tube shielded by a grid out μ-metal to attenuate Earth's magnetic field and electronics for power supply and readout, all housed within a spherical housing made from quartz that can resist the pressure of the ice. The DOMs have a vertical spacing of 17 m and are deployed in depth between 1450 km and 2450 km arranged in strings. There are 86 strings with a horizontal distance of 125 m distributed in triangles such that together the whole detector volume is ≈ 1 km^3 [Col01]. The energy and direction of the original neutrino can be reconstructed and used to search neutrino sources. The PMTs of the DOMs specified to detect light between 300 nm and 650 nm [Aar17]. The wavelength shifting optical module (WOM) has the advantage of being sensitive to UV light. While the differential number of photons per wavelength λ increases proportional to $\propto \lambda^{-2}$ and the effective area scales with the number of photons the module is sensitive for the response on the Cherenkov light is higher. Because the dark count rate does not correlate with the size, the signal-to-noise ratio is lower for a WOM with a DOM of comparable sensitive area size.

There are other projects the WOM can be used for.

It is possible to use its main component to detect light emitted in a scintillator. The surround background tagger is a veto detector to reject the background signal for Search for Hidden Particle (SHiP). SHiP is a fixed target facility on the CERN SPS that investigates the hidden particles such as dark photons, heavy new leptons, light scalars, and axions [Ane15]. The particles decay in a vacuum vessel evacuated at the level of 10^{-6} bar and 50 m long with a cross section of 2.2×5 m^2 in the beginning and 6×11 m^2 in the end [Ahd18]. The whole decay volume is surrounded by liquid scintillator cells, which use the wavelength shifting light guide of the WOM to detect the scintillator photons with photo-sensors [Ane15]. Die verwendung des WOMs im tagger wichtige punkte/eigenschaften des WOMs für ship

This work aims at the reproducibility of the coating process of the light guide and its characterization, such as the variance of the efficiency from different directions and the overall efficiency at all.

In the beginning, it is described how 35 wavelength shifting light guides have been coated.

Twenty-two of them were characterized.

Dependencies of the geometry of the light guide were investigated as well as the difference between light guides produced with polymethylmethacrylate (PMMA) and quartz glass.

For the characterization, a semi-automatized setup was build-up which was able to estimate the capture efficiency in less than four hours.

The setup was tested with a cross-check with an efficiency measuring setup of the Johannes Gutenberg-Universität Mainz.

In the end, the light distribution of one light guide at its cutting edges was investigated in dependency to the point of illumination.

2. Theory

This chapter will describe the concept of the Wavelength Shifting Module (WOM) and its theoretical performance. Afterward, the primary measurement device, the photomultiplier tube (PMT), will be introduced, and later, the theoretical basis for the data reduction will be provided.

2.1 The Wavelength Shifting Optical Module (WOM)

The Wavelength Shifting Optical Module consists of a transparent cylindrical light guide (number 1 in Figure 2.1) coated with a wavelength shifting (WLS) paint (2), which captures photons by internal total reflections within the light guide (3). Photomultiplier tubes (PMTs) on the ends (4) detect the captured light. As a light guide, a transparent rod or tube can be used. In the tube's case, an adiabatic light guide (5) is used to collect the light to a photocathode of a smaller diameter.[1] Together with the readout electronics, all components are housed in a commercial quartz pressure vessel (6).

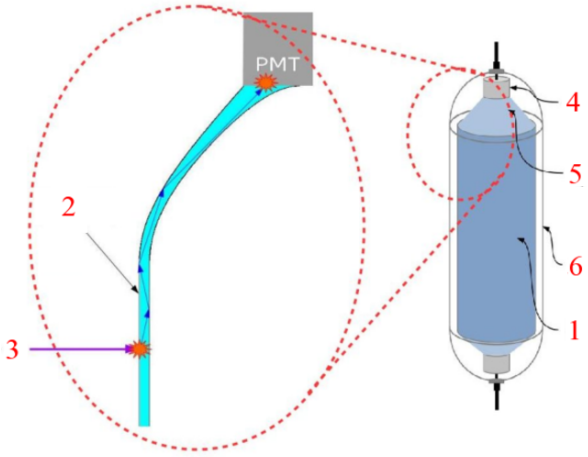


Fig. 2.1: Schematic of a WOM. 1: WLS painted tube 2: WLS paint 3: Absorbed photon 4: PMT and Readout 5: Adiabatic light guide 6: Pressure vessel [Heb14]

UV light is absorbed by the paint and re-emitted with a longer wavelength caused by the Stokes-Shift. The residual energy is transferred to the medium with phononic interactions.[2] The emitted photon is captured by total reflection and propagates to the ends where the PMTs can detect it. The sensitive area of the WOM depends on

[1] The adiabatic light guide has been developed in [Fal14].
[2] For a detailed theoretical explanation about the Stokes-Shift, see [Heb14].

the length and the radius of the cylindrical light guide. The adiabatic light guide concentrates the signal on a smaller photocathode and, therefore, reduces the dark count rate. The signal-to-noise ratio can be increased by increasing the sensitive area. The maximum sensitivity of the photomultiplier used in the DOMs is at a wavelength of 420 nm [Cor03]. The wavelength-shifting paint absorbs UV light with a minimal wavelength of 380 nm [Heb14]. Figure 2.2 shows the absorption and emission spectra of the first component of the paint 1,4-Bis(2-methylstyryl)benzene (Bis-MSB). The second part, p-Terphenyl, absorbs wavelength shorter than 300 nm and re-emits the light in the absorption range of Bis-MSB. The re-emitted light has its maximum at 420 nm, which can be detected more efficiently with PMTs. The WOM is sensitive to wavelengths up to 400 nm. However, for the application of detecting Cherenkov radiation, a higher signal-to-noise ratio can be expected because the number of photons per wavelength is proportional to $\propto \lambda^{-2}$ and the dimension of the photocathode is decoupled from the size of the sensitive area. A practical advantage is that the drilling costs per unit sensitive area are lower for a cylindrical-shaped WOM than for a spherical DOM. In this work, the focus lies on the WLS-coated cylindrical light guide, which will be called WOM from now on. Figure 2.3 shows a WOM prototype with adiabatic light guides illuminated with UV light.

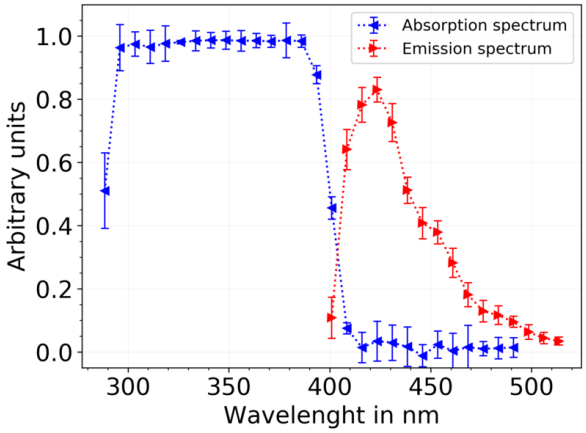


Fig. 2.2: The absorption and emission spectrum of 1,4-Bis(2-methylstyryl)benzene (Bis-MSB) in arbitrary units. Measured in [Heb14].

2.1.1 Theoretical capture efficiency

The efficiency of the WOM depends on the efficiency of its parts.

$$\varepsilon_{\mathrm{WOM}}(z,\Phi) = \varepsilon_{\mathrm{T}}(z,\Phi)\cdot\varepsilon_{\mathrm{CPT}}\cdot\varepsilon_{\mathrm{ALG}}\cdot\varepsilon_{\mathrm{WLS}}\cdot\int_0^\infty d\lambda Q_{\mathrm{PMT}}(\lambda)I_{\mathrm{WLS}}(\lambda)\cdot\int_0^\infty d\lambda A_{\mathrm{WLS}}(\lambda)I_{\mathrm{LS}}(\lambda) \tag{2.1}$$

With ε as the overall efficiency and ε_{T} as the transport efficiency, which depends on the point of illumination. $\varepsilon_{\mathrm{CPT}}$ is the geometrical capture efficiency given by

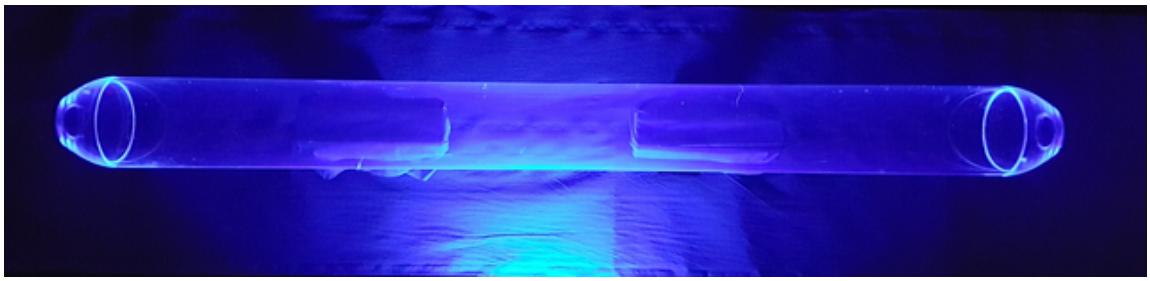


Fig. 2.3: Prototype of a light guide for a wavelength-shifting optical module. The tube's material is PMMA (length: 900mm, diameter: 90mm, wall thickness: 3mm). It is illuminated with UV light.

the angle of total reflection, $\varepsilon_{\mathrm{ALG}}$ the efficiency of the adiabatic light guide, and $\varepsilon_{\mathrm{WLS}}$ is the efficiency of the WLS to convert and re-emit absorbed photons. $\varepsilon_{\mathrm{WLS}}$ combines the efficiency of all conversion processes of the wavelength shift whereby the wavelength dependency is neglected. It represents the ratio of the absorbed and the emitted power by the paint. Q_{PMT} is the wavelength dependent quantum efficiency of the PMT and A_{WLS} the absorption spectrum of the paint. Both spectra are dimensionless I_{WLS} and I_{LS} are the emission spectra of the paint and the light source as probability density functions to emit a photon of a certain wavelength in $\frac{1}{\mathrm{nm}}$. The emission spectra are normalized by their integral to one. Polar coordinates are used to describe the position on the cylindrical surface of the light guide with z as the position at the radial axis with the origin in the WOM's center and Φ as the azimuth angle. In this work, the adiabatic light guide is excluded, and therefore $\varepsilon_{\mathrm{ALG}} = 1$. The product of the efficiencies $\varepsilon_{\mathrm{T}}(z, \Phi)$, $\varepsilon_{\mathrm{CPT}}$ and the probability of absorption and re-emission $\varepsilon_{\mathrm{WLS}} \cdot \int_0^\infty d\lambda A_{\mathrm{WLS}}(\lambda) I_{\mathrm{LS}}(\lambda)$ can be measured directly. For this, the WOM is illuminated at a point (z, Φ) with a light source. Two PMTs cover both ends of the WOM completely, and the number of photoelectrons $\mu^{\mathrm{PMT}}(z, \Phi)$ are detected. These are compared with reference values $\mu_{\mathrm{ref}}^{\mathrm{PMT}}$ measured by illuminating the PMTs directly to get the one-sided efficiencies $\varepsilon^{\mathrm{PMT}}$. To estimate the two-sided efficiency ε, both $\varepsilon^{\mathrm{PMT}}$ are summed together. It is important to consider that the light source has a different emission spectrum I_{LS} than the re-emitting paint I_{WLS}. The response of the PMT for both spectra is different in general. The reference value needs to be corrected by the ratio $\kappa_{\mathrm{corr}}^{\mathrm{PMT}}$ for the different PMT responses. The measured efficiency $\varepsilon(z, \Phi)$ follows as:

$$\varepsilon^{\mathrm{PMT}} = \frac{\mu^{\mathrm{PMT}}}{\mu_{\mathrm{ref}}^{\mathrm{PMT}}} \cdot \kappa_{\mathrm{corr}}^{\mathrm{PMT}} \tag{2.2a}$$

$$\kappa_{\mathrm{corr}}^{\mathrm{PMT}} = \frac{\int_0^\infty d\lambda Q_{\mathrm{PMT}}(\lambda) I_{LED}(\lambda)}{\int_0^\infty d\lambda Q_{\mathrm{PMT}}(\lambda) I_{\mathrm{WLS}}(\lambda)} \tag{2.2b}$$

$$\varepsilon = \sum_{\mathrm{PMT}} \varepsilon^{\mathrm{PMT}} \tag{2.2c}$$

It is assumed that all photons hitting the photocathodes are absorbed or detected. Reflections back into the light guide are neglected. The number of photoelectrons measured at one side μ^{PMT} is the product of the number of emitted photons by the light source N and half the theoretical efficiency of the overall WOM $\varepsilon_{\mathrm{WOM}}$. For the reference value $\mu_{\mathrm{ref}}^{\mathrm{PMT}}$ only the response of the PMT has to be taken into account:

$$\mu^{\mathrm{PMT}} = N \frac{\varepsilon_{\mathrm{WOM}}}{2} \tag{2.3a}$$

$$\mu_{\mathrm{ref}}^{\mathrm{PMT}} = N \int_0^\infty d\lambda Q_{\mathrm{PMT}}(\lambda) I_{\mathrm{LS}}(\lambda) \tag{2.3b}$$

It follows with the equations 2.1 and 2.2 that:

$$\varepsilon(z, \Phi) = \varepsilon_{\mathrm{T}}(z, \Phi) \cdot \varepsilon_{\mathrm{CPT}} \cdot \varepsilon_{\mathrm{WLS}} \cdot \int_0^\infty d\lambda A_{\mathrm{WLS}}(\lambda) I_{\mathrm{LS}}(\lambda) \tag{2.4}$$

By using a light source emitting only in the range of the plateau of the absorption spectrum S_{WLS} of the paint (see Figure 2.2), the integral can be simplified to an absorption coefficient a_{WLS} which depends on the paint thickness and absorption length. The above Equation 2.4 can be rewritten as:

$$\varepsilon(z, \Phi) = \varepsilon_{\mathrm{T}}(z, \Phi) \cdot \varepsilon_{\mathrm{CPT}} \cdot \varepsilon_{\mathrm{WLS}} \cdot a_{\mathrm{WLS}} \tag{2.5}$$

The geometrical capture efficiency $\varepsilon_{\mathrm{CPT}}$ is given by the angle of total reflection depending on the optical densities of the surrounding medium n_{M} and the material of the cylindrical light guide n_{LG}. After the absorption of the short wavelength photon a shifted photon will be re-emitted isotropically by the wavelength shifting molecule. The excited molecule can be seen as a point source embedded in an approximately flat plate with a finite distance to the surface. The angle of total reflection confines the solid angle Ω_{C} in which the re-emitted photons will be captured in the optical dense material with a density of n_{LG}. $\varepsilon_{\mathrm{CPT}}$ is the ratio between Ω_{C} and 4π [Heb14].

The angle of total reflection follows from Snell's Law:

$$\omega = \arcsin\left(\frac{n_{\mathrm{M}}}{n_{\mathrm{LG}}}\right) \tag{2.6a}$$

$$\Omega_{\mathrm{C}} = \int_0^{2\pi} d\phi \int_{\omega}^{\pi-\omega} d\theta \sin(\theta) \tag{2.6b}$$

$$\varepsilon_{\mathrm{CPT}} = \frac{\Omega_{\mathrm{C}}}{4\pi} = \cos(\omega) \tag{2.6c}$$

The geometrical capture efficiencies for the materials used in this work are listed in Table 2.1.

Material	n_{LG}	ω in $^\circ$	$\varepsilon_{\mathrm{CPT}}$ in %
PMMA	1.49	42.2	74.1
quartz	1.46	43.2	72.9

Tab. 2.1: The geometrical capture efficiencies $\varepsilon_{\mathrm{CPT}}$ of PMMA and quartz glas for the case of air $n_M \approx 1$ as surrounding medium. The optical densities are taken from [Kuc96].

Which also defines the theoretical limit for the efficiency of the WOM. If the paint's absorption length compared with its thickness is negligible, a_{WLS} can be assumed to be one as well. The left free parameter estimated by this measurement is the product of the transport efficiency ε_{T} and the conversion efficiency $\varepsilon_{\mathrm{WLS}}$ of the paint.

2.2 Photomultiplier tube

The photomultiplier tube (PMT) is a vacuum tube used to measure single photons. It generates a measurable current by 'multiplying' the primary photoelectrons generated with the photoelectric effect a million times. The primary electron from the cathode material's surface is accelerated with an electrical field to the first dynode. When hitting a dynode, it generates electrons in the order of $O(1)$ to $O(10)$ other photoelectrons, accelerated to the next dynode. Depending on the number of dynodes and the supplied voltage, the PMT can have an average gain G up to the order of $O(10^9)$. Typical gains are between 10^5 and 10^7. The multiplication and the trajectories are affected by external magnetic fields. Magnetic protection with an μ-metal shielding is recommended [Kol16]. The generated charge q on the anode discharge over a resistance, which results in a voltage. This can be measured over time, which will be called an event or waveform from now on. q is estimated by the integration of those events. The distribution $T(q)$ has two contributions. The first is the background

noise, which is denoted as $B(q)$ and is always present even without illumination. It is the result of electronic noise and dark counts. The second is the signal $S_\mu(q)$ depending on the average number of primary photoelectrons μ. Both are convoluted to the total charge distribution [Sal17]:

$$T_\mu(q) = (B * S_\mu)(q) \tag{2.7}$$

For independent random distributions, the averages and variances are additive:

$$\langle T \rangle_q = \langle B \rangle_q + \langle S_\mu \rangle_q \tag{2.8a}$$
$$V[T] = V[B] + V[S_\mu] \tag{2.8b}$$

The background noise $B(q)$ must be determined by an independent measurement collecting events without the illumination of the PMT. The signal can be described as:

$$S_\mu(q) = \sum_{p=0}^{\infty} \Psi_p(q) P_\mu(p) \tag{2.9}$$

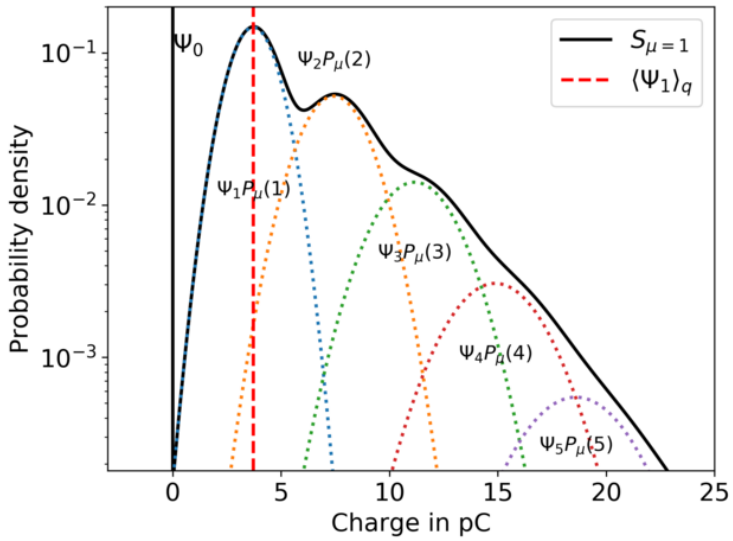


Fig. 2.4: Theoretical charge distribution of the signal S for the case of $\mu = 1$, $\langle\Psi_1\rangle_q = 3.72$ pC and $V[S_\mu] = 1$ pC^2.

$P_\mu(p)$ is the discrete probability of having p photoelectrons recorded in an event distributed around a Poisson average μ and $\Psi_p(q)$ the continuous charge distribution for p photoelectrons. Figure 2.4 shows the theoretical charge distribution given by Equation 2.9. It is assumed that the distribution of the single photoelectron peak Ψ_1

is Gaussian normalized by its integral to one. The pedestal Ψ_0 is a delta function. The average of $\Psi_p(q)$ is $p \cdot \langle \Psi_1 \rangle_q$ and the variance is $p \cdot V[\Psi_1]$ with $\langle \Psi_1 \rangle_q$ and $V[\Psi_1]$ as the average and variance of the single photoelectron peak whereby $\langle \Psi_1 \rangle_q$ depends linearly on the gain G of the PMT. Both can be determined straightforwardly if the average number of photoelectrons per event μ is known. With the above equations 2.8 and 2.9 it follows:

$$\langle \Psi_1 \rangle_q = \frac{\langle T_\mu \rangle_q - \langle B \rangle_q}{\mu} \tag{2.10a}$$

$$V[\Psi_1] = \frac{V[T_\mu] - V[B] - \langle \Psi_1 \rangle_q^2 * V[S_1]}{\mu} \tag{2.10b}$$

From this, the gain G of the PMT is calculated with the elementary charge e:

$$G = \frac{\langle \Psi_1 \rangle_q}{e} \tag{2.11}$$

The errors of equations 2.10 are discussed in [Sal17].

2.2.1 Counting photons - The model-independent approach

The approach to counting photoelectrons without a model of the response of the PMT is described and investigated in [Sal17]. To estimate the average number of photoelectrons detected by the PMT per event, the probability for no photoelectrons $P_\mu(0)$ needs to be determined. The appearance of light can be seen as Poisson distributed.

$$P_\mu(k) = \frac{\mu^k}{k!} e^{-\mu} \tag{2.12}$$

The average number μ follows analytically from the Poisson Equation 2.12 for the case of $k = 0$. Events without photoelectrons are called pedestal events from now on. The probability $P_\mu(0)$ is the ratio between the number of those events $N_{k=0}$ and the overall number of collected events N_{total}. From Equation 2.12 it follows for μ:

$$\mu = \log\left(\frac{N_{total}}{N_{k=0}}\right) \tag{2.13}$$

The real photomultiplier tube cannot amplify all the photoelectrons with its characteristic gain because a large variety of sub-optimal trajectories exist for the photoelectrons. This increases the part T_μ with fewer charges [Sal17]. Furthermore, the pedestal is penetrated by the different distributions $\Psi_p(q)$, especially from the

single photoelectron peak $\Psi_1(q)$. To take this and the background noise into account, the pedestal events of the measurement $T_\mu(q)$ are compared with the pedestal of a background dataset $B(q)$. This second data is measured under the same conditions but without illumination. For comparison the f-quantile q_c of B will be estimated. This cutoff is chosen such that the number of under-amplified photoelectrons until q can be neglected. In this work, the fraction was $f = 30\%$.

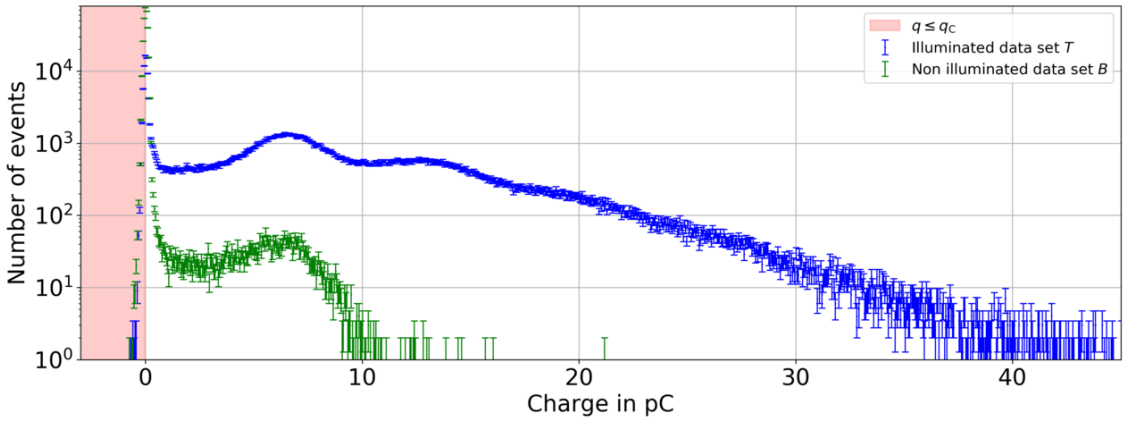


Fig. 2.5: Blue: The dataset T of an illuminated XP2020 with (1.595 ± 0.015) photoelectrons in average per event. Green: The background dataset B without illumination. Red shaded: Region below cutoff q_c where 30% of the events of B fall below.

The number of events A_T in T until this cutoff is counted. The total number of pedestal events is therefore $N_{k=0} = {}^{A_T}/_{f}$. Together with Equation 2.13, the Poisson average can be calculated with:

$$\mu = \log\left(\frac{N_B}{A_T}\right) \tag{2.14}$$

Whereby $N_B = f \cdot N_{total}$ is the actual number of events falling below q_c. From Gaussian error propagation, it follows for the statistical error of μ:

$$\Delta\mu = \sqrt{A_T^{-1} + N_B^{-1}} \tag{2.15}$$

It was assumed that the statistical error of the number of events until the cutoff A_T and N_B is simply the square root of those. For a more detailed explanation of the model-independent approach, see [Sal17].

3. Coating

An important part of WOM production is the coating of the light guide with the wavelength-shifting paint. It is assumed that the conditions of the coating have a major impact on the latter efficiency. The commercial ND-DC Dip Coater from the company Nadetech Innovations S.L. has been used for coating. The Dip Coater had a maximal displacement of 300 mm and a maximal load of 1 kg. The maximal displacement velocity was 150 $\frac{\text{mm}}{\text{min}}$. It was programmable in terms of immersion height, velocity, and interim stops by the manufacturer's software. Figure 3.1 shows the Dip Coater with a mounted WOM. The paint was contained in a cylindrical aluminum vessel.[1] In general, the WOMs were coated on the outside only. In the case of tubes, caps were needed to close the inner part. The caps were made out of aluminum and were kept together with three threaded rods (see Figure 3.1). The rods had a minimal distance of 7 mm to the light guide's surface. [2]

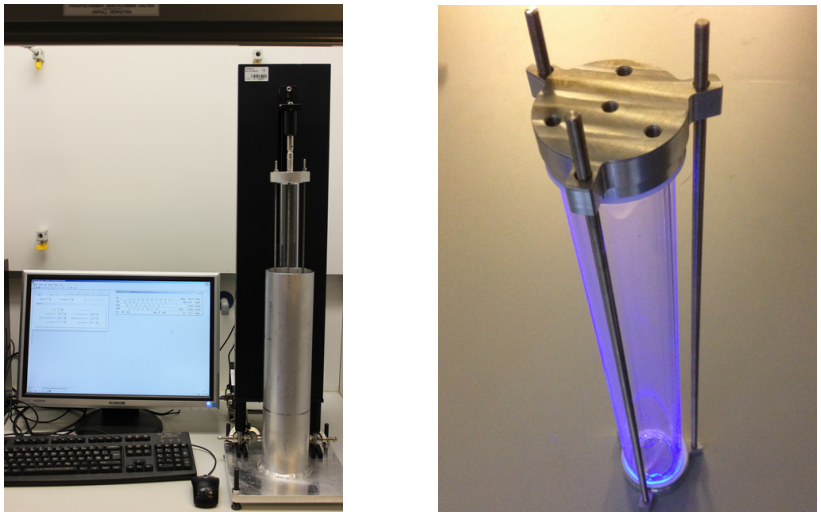


Fig. 3.1: Left: The commercial ND-DC Dip Coater from Nadetech Innovations S.L. with mounted WOM and the paint containing aluminum vessel. Right: The caps mounted at a PMMA-WOM.

[1] Inner diameter: 90 mm, wall thickness: 5 mm, Height: 430 mm
[2] For technical drawing see the appendix Section A.4.

3.1 Wavelength shifting paint

The wavelength-shifting paint was developed in the work of Hebecker [Heb14]. Toluene works as a solvent. The paint has two different wavelength-shifting molecules. The 1,4-Bis(2-methyl styryl)benzene (Bis-MSB) absorbs mainly in the region from 300 nm to 400 nm. Its absorption and emission spectrum can be seen in Figure 2.2. To absorb wavelengths shorter than 300 nm p-Terphenyl is used. The second shifting molecule re-emits in the absorption range of the Bis-MSB. It is assumed that the sequential conversion with both molecules and the single conversion with Bis-MSB can only be described with the constant efficiency $\varepsilon_{\mathrm{WLS}}$. Ethyl-methacrylate copolymer (EMC) works as the matrix for the wavelength-shifting molecules. The composition of the paint developed in [Heb14] is shown in Table 3.1 as the target concentration.

Component	Target mass concentration $[10^{-3}]$	Actual mass concentration $[10^{-3}]$	Delivered from
Toluene	773	784.79 ± 0.04	VWR Chemicals
EMC	222	211.42 ± 0.02	Kremer-Pigmente
Bis-MSB	1.3	1.273 ± 0.006	Sigma-Aldrich
p-Terphenyl	2.6	2.521 ± 0.007	Sigma-Aldrich

Tab. 3.1: Target and measured mass concentrations in per mill of the wavelength shifting paint used in this work. The EMC had a impurity of ≤ 2 % the others were ≥ 99 % pure.

2319 g of the paint was produced in total. The paint was made in seven different batches and mixed. The toluene was heated with a magnetic stirrer in an 800 mL beaker. The heater was set to 150 °C.[3] The stirrer was set to 400 min^{-1} at the beginning. In the next step, the EMC was solved in the toluene. To avoid agglutination, it was added in 5 g to 10 g portions to the toluene. The next portion of EMC was added after the previous was solved completely. The stirrer's velocity was increased every time the vortex in the solution disappeared due to the rising viscosity [Heb18]. The vessel the chemicals were weighed in before and after the addition.[4] The actual added amount of the component was estimated with the mass difference of the vessel. The chemicals were tipped into the solution only.[5] After the last component was added, the paint was stirred an additional 5 min. In the end, the

[3]This temperature is empirically tested in [Heb14], and it is assumed that the toluene will not reach its boiling temperature of ≈ 111 °C [Kuc96] during the production.

[4]KB360-3N from the company Kern & Sohn GmbH, Serial number: W1500370 with a maximal weight of 360 g. The scale was adjusted with F2 calibration weights from the same manufacturer before each usage.

[5]The use of a spoon-like tool to add the component to the solution would have compromised the mass difference.

total amount of paint was measured by weighing.[6] It was observed that 5 % to 9 % of the toluene vaporized during the production. The lost toluene was added back to the solution to approach the target concentration. The mass concentrations of the mixed paint are shown in Table 3.1.

3.2 Coating speed

The thickness of the coated paint depends on the withdrawal velocity of the substrate. If the paint's absorption length is not negligible against its thickness, only a fraction of the ultraviolet light will be converted, and the rest will be lost. The dependency of the paint thickness on the coating speed was investigated in [Heb14] and is shown in Figure 3.2. The thickness saturates at a speed of $\approx 40\ \frac{\text{mm}}{\text{min}}$. It can be concluded that $150\ \frac{\text{mm}}{\text{min}}$ is enough to achieve the maximal layer thickness and, therefore, maximal available absorption. Nevertheless, to guarantee repeatability, the Dip Coater's displacement speed should be chosen such that the coating speed is always the same independent of the displacement of the paint due to the immersed WOM. The new level of a liquid displaced by a body of constant cross-section which is immersed up to the ground can easily be derived from the ratio g of the liquid total surface without A_{paint} and with $A_{\text{displaced}}$ displacing body. The effective coating speed ν_{eff} follows from the same thoughts:

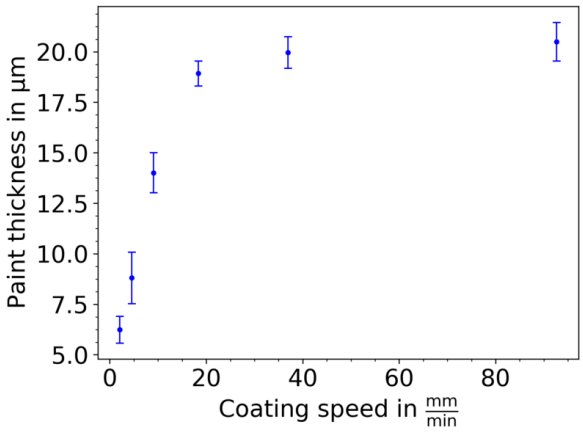


Fig. 3.2: Dependency of the paint thickness on the withdraw velocity estimated in [Heb14].

[6]The paint was produced in the Institut für Physik, Room 2.113, Newtonstr. 15, DE-12489 Berlin, with the usage of a fume cupboard.

$$h_{\text{eff}} = h \cdot g \tag{3.1a}$$

$$\nu_{\text{eff}} = \nu \cdot g \tag{3.1b}$$

$$g = \frac{d_{\text{vessel}}^2}{d_{\text{vessel}}^2 - d_{\text{WOM}}^2 - 3d_{\text{rod}}^2} \tag{3.1c}$$

Whereby h is the immersion depth with respect to the Dip Coater, h_{eff} is the actual immersion depth due to the displacement with respect to the new paint level, ν the displacement speed of the Coater, d_{vessel}, d_{WOM} and d_{rod} the diameter of the paints vessel, the WOM and the three threaded rods. In general ν was chosen such that $\nu_{\text{eff}} = 150\ \frac{\text{mm}}{\text{min}}$.

3.3 Coating process

For dip coating, the cleaning procedure is important. Any dust particle or lipid layer on the surface can prevent a full moistening of the WOM's surface. As a consequence, parts of the WOM will left uncoated and, therefore, insensitive to light. As a cleaning cloth, 'Kimtech SCIENCE Labortücher' has been used, which is fuzz-free and made for cleaning optical apertures. To clean PMMA isopropanol[7] was used and toluene[8] for quartz glass. The inner side of the PMMA tubes has been cleaned from dust as well and was formed into a ball, moistened with isopropanol or toluene, and pushed through. The used isopropanol left a thin layer of unknown substances on the PMMA surface. Dry cleaning clothes were used to remove these layers. Because of the electrostatic on the tube's surface, it had to be stored for an hour. After that, air pressure was used to remove the dust from the surface. In the case the cylindrical light guide was made of PMMA, it was weighted with a scale.[9] In the next step, the caps (see Figure 3.3) were mounted on the tube and the WOM on the dip coater (see Figure 3.1). To seal the tubes, commercial seals of the company 'Salaske & Bierau GbR IR Dichtungstechnik' have been used[10]. The material was Polytetrafluoroethylene (PTFE). To get the tubes tightened and avoid having paint contact with the ends, it was observed that the needed torque was enough to compress the PMMA tubes by 1 mm concerning its length of 300 mm.

The effective immersion speed $\nu_{\text{eff}}^{\text{i}}$ was chosen for all WOMs as 80 $\frac{\text{mm}}{\text{min}}$ and the effective withdraw speed $\nu_{\text{eff}}^{\text{w}}$ as 150 $\frac{\text{mm}}{\text{min}}$. Between immersion and withdrawal, the

[7] Technically pure (purity $\geq$ 98 %)
[8] Analytically pure (purity $\geq$ 99.8 %)
[9] It was used the same scale as in Section 3.1.
[10] Outer diameter: 52 mm, inner diameter: 31 mm and the thickness: 2 mm



Fig. 3.3: The aluminum caps to close the tubes.

tube remained in the paint for 120s. In the end, the WOM was unmounted from the Dip Coater and hung into a fume hood for two days minimum.

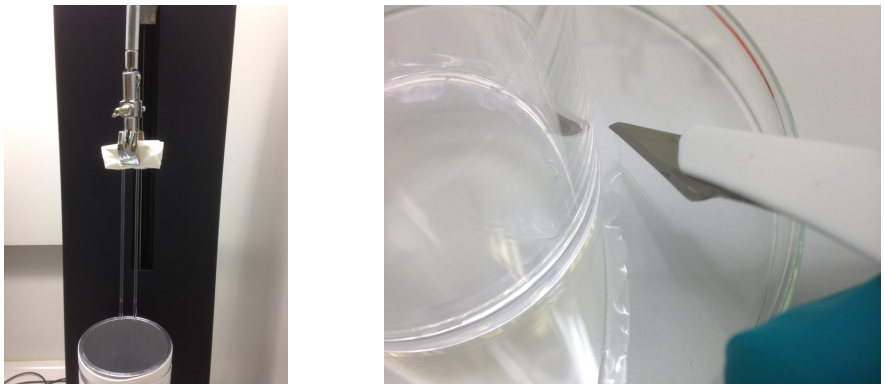


Fig. 3.4: Left: Coating of 3 mm rods. Right: Remove surplus paint from the lower end of the tube. A culture dish was used underground.

After drying, the tubes were unmounted from the caps. On the lower end, the paint needed to be removed with a scalpel (see the right on Figure 3.4). The rods were clamped pairwise on a stand clamp (see Figure 3.4). To prevent the rods from being coated on the lower end, Teflon tape was wrapped around the end. Nevertheless, after drying and removing the Teflon tape, it was observed that the paint came into contact with the rods' ends. Those were removed with cleaning clothes and isopropanol. In the end, all PMMA-made WOMs were weighted again. All WOMs were coated at a room temperature between 22 ° C and 24 ° C and relative humidity between 30 % and 45 %.

3.4 Paint thickness

During the production process of the WOM described in 3.3, it was possible to weigh the rods or tubes before and after the coating. To calculate the paint thickness, the mass difference $\Delta m = m_{\text{after}} - m_{\text{before}}$ was calculated with the weight m_{before} of the light guide before coating and after m_{after}. The paint thickness follows with:

$$d_{\text{paint}} = \frac{m_{\text{after}} - m_{\text{before}}}{A_{\text{WOM}} \cdot \rho_{\text{EMC}}} \tag{3.2}$$

Only the weight of the PMMA WOMs could be estimated because the weight of the glass tubes was out of the range of the scale used. Figure 3.5 shows the area densities of the WOMs.

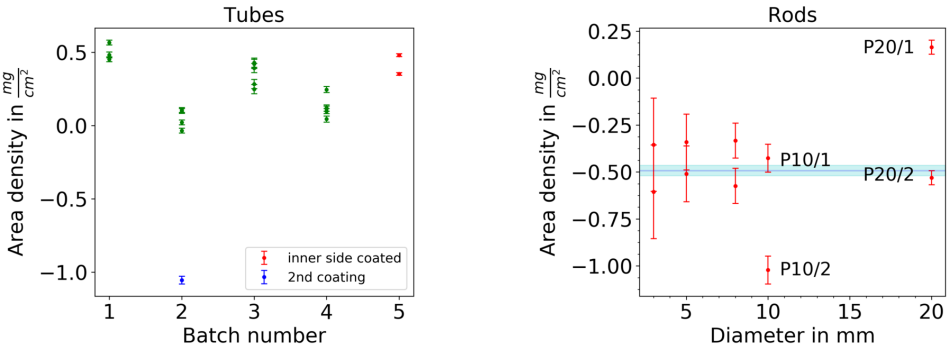


Fig. 3.5: The estimated area density of the paint for the coated PMMA tubes and rods. Left: The area density of all tubes. The tubes are coated in up to six different batches. Right: The area density of the different rods is dependency of the diameter.

In the left figure, the area density fluctuates between the different series. The area densities of the rods in the right of Figure 3.5 are negative. With except the two outliers, the area density can be seen as reproducible around the value:

$$(-0.49 \pm 0.03)\frac{\text{mg}}{\text{cm}^2} \tag{3.3}$$

It was observed that the PMMA was electrostatically charged in a way such that it was able to compromise the weighing measurement. This could be seen by holding the tube above the weighing plate without contact, which leads to a negative weight. In the case of the measurement method of the scale, which is the strain gauge, the electrostatic charge only attracts the plate (made out of plastic). In this case, the results are trustworthy, and as a consequence, it could not get reproducible paint thickness. The second possibility is that the scale used a capacitive sensor affected by the electromagnetic charge on the WOM's surface. The tubes' values have no validity. Before weighing the rods, it was considered that the cleaning process charges them, and they were weighed with a delay of a day.[11] The scale did not show any effect of electromagnetic charge such that it can be assumed that the estimated area densities of the rods are not compromised (see Figure 3.5). With the density of the EMC, the averaged paint thickness can be determined as (-0.57 ± 0.03) μm. From

[11] Another possibility would have been to remove the electrostatic charge with commercial deionization systems.

this, more PMMA of the rods has been solved than paint has been coated on the WOM. Furthermore, the WOM is now tapered because the lower part of the WOM had 5.75 min longer contact with the paint than the upper part. The loss in radius of (-0.57 ± 0.03) μm is therefore only in average. To prevent the lower part of the rods from being coated on the polished ends, they have been sealed with Teflon tape. The tape was not wrapped around uniformly. It could be that this had an effect on the paint flowing downwards during the drying process, which explains the two outliers, P10/2 and P20/1.
A possibility to estimate the paint thickness would be by weighing quartz WOMs instead because quartz is not solvable. A scale with a maximal load of 1 kg and a mg resolution would be necessary.[12]

3.5 Seals

As described in Section 3.3, PTFE seals have been used to seal the caps.[13] The needed force was high enough to compress the PMMA tubes significantly with a relative contraction $\Delta l/l$ of $1/300$. If the steel-made threaded rods' elongation is neglected, the needed torque can be estimated for an M5 thread. The contracting force was between 3.7 kN and 6.7 kN, which can be generated from a torque of 11 Nm to 20 Nm per nut.[14] It is likely that the torque will vary from nut to nut, even for one tube, which will lead to perpendicular forces of significant size. If the compression and the following extension of the paint after unmounting the caps affect its quality, the reproducibility will be compromised. At those forces, it is also conceivable that a quartz glass tube is damaged if the torque is not controlled. To avoid perpendicular forces, a torque wrench is recommended. This will help to make the tubes reproducible and efficient on the caps.
To avoid those forces, the second type of seal has been tested. The material was expanded PTFE with the commercial name of 'MuliTex ePTFE' of the company 'Thoenes Dichtungstechnik GmbH.' The material is much softer and, therefore, a more promising candidate. It has been observed that the seals tightened the caps satisfactorily without using significant torque. Nevertheless, it was only practical if the seal had the exact same outer diameter as the WOM; otherwise, it could not be removed due to additional deformation of the seal itself.

[12] Another possibility would be to use a scale with a counterweight.
[13] Delivered from IR Dichtungstechnik Ingeborg Salaske & Randolf Bierau GbR.
[14] For the exact derivation see appendix A.3.

4. Efficiency measurement setup

An automatic setup was built to estimate the capture efficiency. The setup has a light source mounted on an automated driveable platform which can be moved in three axes and rotated in two angles to illuminate the WOM from different directions, two photomultiplier tubes[1] to detect the photons captured by the WOM's ends, one data acquisition unit[2] (ADQ) to measure the waveforms of the PMTs and one pulse generator[3] to supply the light source and to trigger the ADQ. The data is collected with a coincidence measurement to reduce background noise. A darkening box covers the setup to screen it from ambient light. The PMTs have a circular sensitive area with a diameter of ≈ 50 mm, and the largest WOM has a diameter of 45 mm. Figure 4.1 shows a schematic and a picture of the setup. The PMTs are supplied with 2200 V for the left PMT and 2300 V for the right PMT with a high voltage supply[4] to generate a gain of 2×10^7 and the light source is a LED with a wavelength of (375 ± 5) nm. An additional 50 Ω resistance is mounted next to the LED, which works as a dummy light source to simulate the noise due to the cables' electromagnetic cross-talk. The LED and the dummy are pulsed with the same pulses sequentially. Coincident with those pulses, the ADQ was triggered. The trigger pulses were generated at two different channels and had to be merged with a logic unit.[5] After merging, it is attenuated by 20.5 dB to reduce cross-talk between the trigger channel of the acquisition unit and its signal collecting channels. The events with pulsed dummy are those for the background data B, which the dataset T with illuminated WOM will be compared with. This data collection method guarantees that all time-varying disturbances are represented in B as they occur in T. The time period the pulse generator operates is 10 μs, and the pulse to supply the LED follows the dummy pulse after 3 μs. This asymmetry allows the separation of the two datasets after the data collection. For every single measurement, 3×10^5 events are collected for each dataset. The high voltage supply and the logic unit are NIM modules, which are supplied by a NIM crate.[6] After data collection, the data is

[1] XP2020 from Philips, SN23890 (left PMT), SN23998 (right PMT), Bases: Cern EP-Type 4238 SN4122 (left PMT), SN053 (right PMT) The PMTs are protected from external magnetic fields with a μ-metal shielding.

[2] ADQ14DC-4C-VG-USB from SP Devices, P/N: 320-224-024-C, S/N: SPP-05032, inventory number: 1908526

[3] Digital delay generator from Stanford Research Systems, Model DG645, SN005665, inventory number: 100 100 190 8434

[4] High Voltage Power supply from CAEN, Model 126, SN516 (left PMT), SN66 (right PMT)

[5] 3-Fold-Logic unit from CAEN Model N405 SN14

[6] NIM: Nuclear Instrumentation Module NIM crate from 'Wiener Stromversorgung,' inventory number: 65-4027

sent from the ADQ to the computer. Each event has a timestamp[7] which is used to separate the background events B from the events of T. The data is directly used to calculate the measured number of photoelectrons for both PMTs because the amount of raw data is up to 1 GB for each dataset of waveforms. Only the resulting number of photoelectrons has been saved. Coaxial cables with LEMO connectors have been used to supply the light source and read out the signals. Only the ADQ needed SubMiniature version A connectors (SMA). To determine the WOM efficiency, the current intensity of the light source needs to be estimated by directly illuminating the PMTs. For this, the PMTs can be rotated so they do not face the WOM's holding structure. To guarantee repeatability, the rotations of the PMTs are locked by holding the structure itself at 90 °. To have optical contact, the PMTs are covered with vulcanized elastic transparent gel[8] with the optical density of 1.404 [AG99]. The WOM's ends are pressed against the photocathodes by translating the left PMT with a linear stage at its axis.[9]

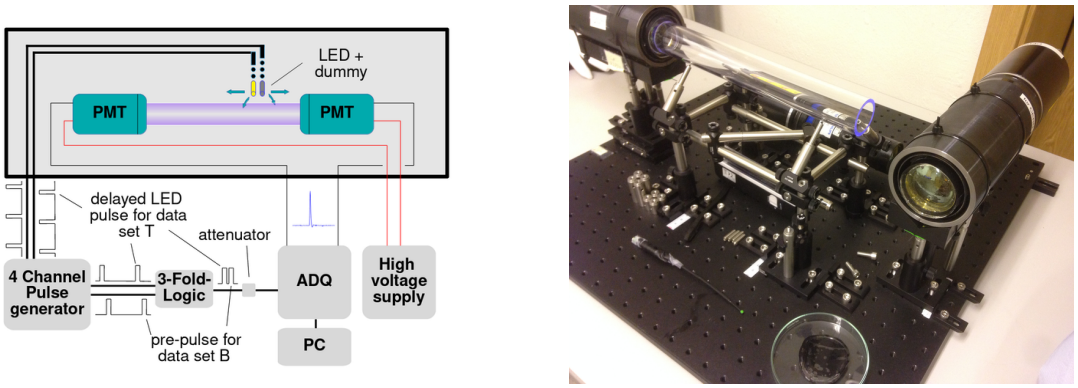


Fig. 4.1: Left: Schematic of the setup. Right: Picture of the setup outside the darkening box.

4.1 Five Axis Scanner

The photomultipliers have to be screened from environmental light. A manual movement of the light source is, therefore, not possible as long as the PMTs are supplied. An automatic drive system is used to move the light source, granting reproducibility and allowing two-dimensional surface scans with a resolution less than 0.3 mm[2]. The drive system was delivered from the company Schneider Electric and built by the DESY intern workshop.[10] The unit receives commands via an intranet

[7] One ADQ time unit is 0.25 ns step.
[8] Section A.2 describes how the PMTs are prepared.
[9] The holding structure consists of optical assemblies from the company Thorlabs.
[10] The engine control unit and its application programming interface (API) are engineered and written by the workshop in DESY Zeuthen.

connection. The application programming interface (API) is a Python package and can be imported into the measurement program. The physical available volume is $1070 \times 680 \times 580$ mm^3 and the maximal load to guarantee the spatial repeatability of 50 μm is 15 kg. The Scanner with its three movable axes can be seen on the left figure of 4.2. On the right, the two mounted rotation stages[11], which allows a rotation of the light source around the two drawn axes.

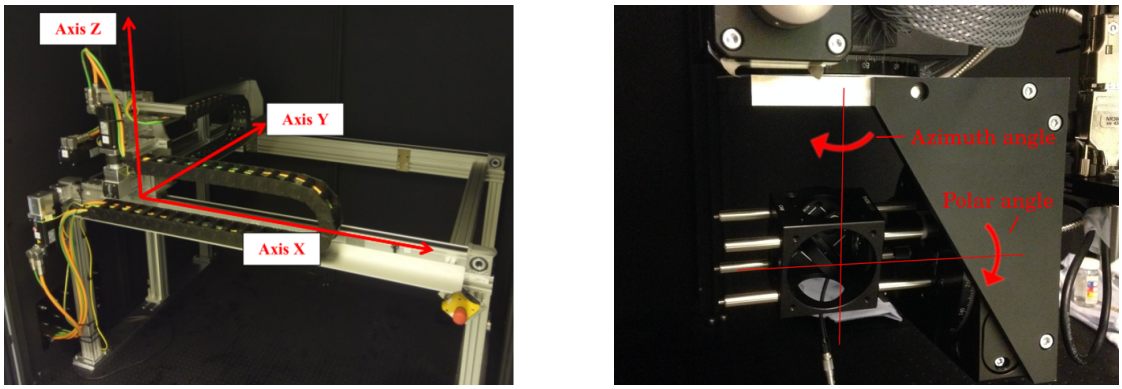


Fig. 4.2: Left: Automatic moveable platform with its three axes in the darkening box. Right: The two rotational stages mounted on the platform with a drawn rotation axis.

4.1.1 Control algorithm

The light source has to be moved automatically over thousands of different points in different directions over a curved surface. Finding the correct trajectory in the five-dimensional parameter space is a nontrivial task. Depending on the situation, it is better to move the z-axis before moving on the XY-plane and the other way around. The question of when the orientation of the light source can be changed makes this more challenging. To avoid the necessity of listing all special cases for different relative positions to the equipment and the relative positions between new and old coordinates and defining trajectories for each case, a more general solution was chosen. The Scanner has a different set of predefined trajectories tested one after another in terms of expected collisions. If no trajectory is found without collision, the program stops the whole measurement.
Each surface that can be scanned is defined by a new coordinate system. The digital module position, which shall be scanned, defines the origin of the new coordinate system. The coordinate system chosen describes the curvature of the investigating module. For a WOM, cylinder coordinates are the best choice. In that way, each

[11] 360 Degree Continuous Rotation Stage with Stepper Motor Actuator from Thorlabs©. The engine control unit is the apt Precision Motion Controller from Thorlabs, Model: BSC203, SN70859020, inventory number: 1908278

object in the volume of the Five Axis Scanner is defined virtually as a coordinate system that can calculate the Scanner's new five-dimensional coordinates p_5 if a tuple x_2 of two-dimensional coordinates and the distance d is given. It is now possible to find the distance between the Scanner's coordinate and the objects by a minimizing process. For this, the distance d_n between the Scanner's future coordinates and the nearest point on the surface is estimated by varying the two-dimensional parameter x_2. The point p_5 is a legal position if the minimal distance d_n is larger than a defined minimal distance d_min. To test a trajectory t from p_s to p_e over the points p_i the minimal distance $d_\mathrm{n}(s)$ will be estimated for each 30 mm. If one point $t(i)$ is found where $d_\mathrm{min} \geq d_\mathrm{n}$, the trajectory is discarded, and the next trajectory will be tried. Only if a trajectory is expected to be safe will the scanner do its movements; otherwise, the test function returns an exception, which stops the whole measurement process, and the Scanner is switched off.

4.2 Light source

As a light source to illuminate the WOM, a (375 ± 5) nm LED[12] was used. The LED is shown in Figure 4.3. It is connected to an acrylic glass fiber of ≈ 10 cm length. To avoid misdirected light, it was covered with black tape. To reduce the opening angle δ, the fiber's end was covered with a shrink-on tube, which stands out. It worked as a pinhole with the distance of (10 ± 2) mm (see schematic of Figure 4.4). The light source's opening angle has been estimated, and it was investigated in terms of stability.

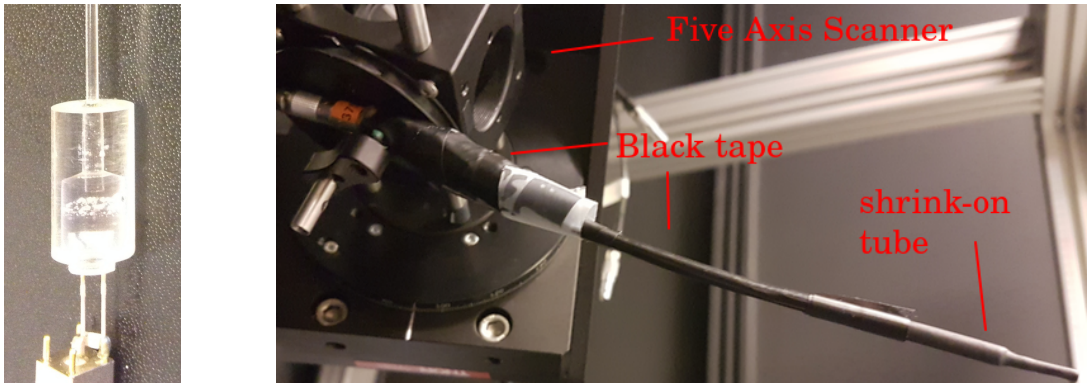


Fig. 4.3: Light source. Left: the uncovered LED is embedded within an acrylic glass housing and connected to fiber. Right: The LED is covered with black tape and mounted on the Five Axis Scanner.

[12]XSL-375-3E from Roithner Lasertechnik GmbH

4.2.1 Opening angle

To estimate the opening angle of the light source, one PMT was covered with aluminum foil. The foil had a hole in the center of the PMT's photocathode punctured by a needle. The diameter of the hole was smaller than 0.5 mm. Four two-dimensional scans at different distances l were done. Each scan had a spatial resolution of 0.3 mm and covered an area of (6×6) mm^2. Figure 4.5 shows the scans with 4 mm and 19 mm distance between light source and covering mask. As a cross-check, the sum Σ_{scan} can be used to check the light intensity drift between the measurements. As a measure for the light spot's diameter, the widest location of the region with the sum $\Sigma_{68\%}$ with 68% of Σ_{scan} is taken. The resolution of 0.3 mm is used as an error. Table 4.1 shows the results of the scans.

l in mm	Σ_{scan} in p.e.	$d_{\leq 68\%}$ in mm
4 ± 1	24.4 ± 0.2	1.5 ± 0.3
6.5 ± 1	24.4 ± 0.2	1.6 ± 0.3
9 ± 1	24.2 ± 0.2	1.9 ± 0.3
19 ± 1	23.3 ± 0.2	3.5 ± 0.3

Tab. 4.1: Results of the measurements of the light cone diameter $d_{\leq 68\%}$ in dependency of different distances l.

The intensity was constant within the measurement. The light spot in the scan for 19 mm is not fully covered by the scanned area, which can explain the deviation of Σ_{scan}. A linear regression was done with the estimated values for $d_{\leq 68\%}$. The regression and the data are plotted on the left of Figure 4.6. It can, therefore, be described with the following:

$$d_{\leq 68\%}(l) = a + b \cdot l \tag{4.1a}$$

$$a = (0.78 \pm 0.17) \text{ mm} \tag{4.1b}$$

$$b = (0.14 \pm 0.01) \tag{4.1c}$$

From geometry, it follows (see Figure 4.4) that the half opening angle δ can be determined with the inverse tangent of the half slope b:

$$\delta = 2 \cdot \arctan\left(\frac{b}{2}\right) = (8.0 \pm 0.8)^\circ \tag{4.2}$$

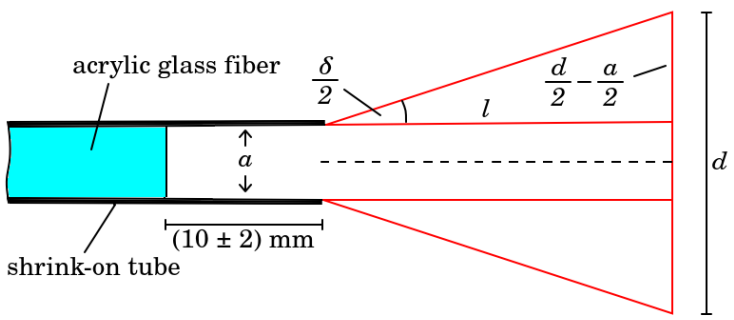


Fig. 4.4: Schematic of the end of the acrylic glass fiber. The end of the fiber is covered with a shrink-on tube, which is (10 ± 2) mm longer. The opening has the diameter a.

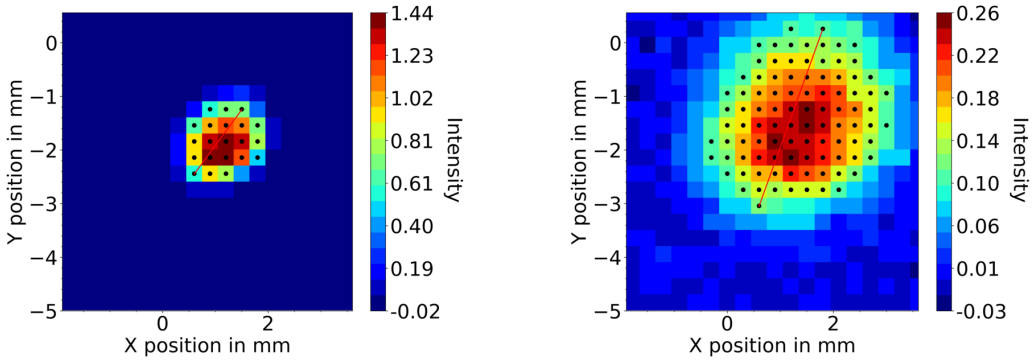


Fig. 4.5: Lightcone scans with 4 mm (left) and 19 mm (right) distance. The black dotted area represents the region $\Sigma_{68\%}$ of 68% of the collected intensity Σ_{scan}. The red line represents the diameter at the widest location.

4.2.2 Stability

To estimate the stability of the light source, the PMTs have been illuminated over 285 h in total over 20 measurements of varying length. The light intensity was measured every minute with 3×10^5 events. Figure 4.7 shows the result of one representative measurement over 18 h. It can be seen that the light source drifts linearly with a constant slope over up to 5 h. This can be considered by taking two reference measurements before and after the scan of the WOM. At hours 6.3, 14.0, 15.2, and 15.5, fast changes in intensity occurred. The right of Figure 4.6 shows the distribution of the amplitude of all 21 jumps observed in 285 h. The average charge is 2 % with a standard deviation of 12 %. The jump with maximal observed amplitude is -30%.

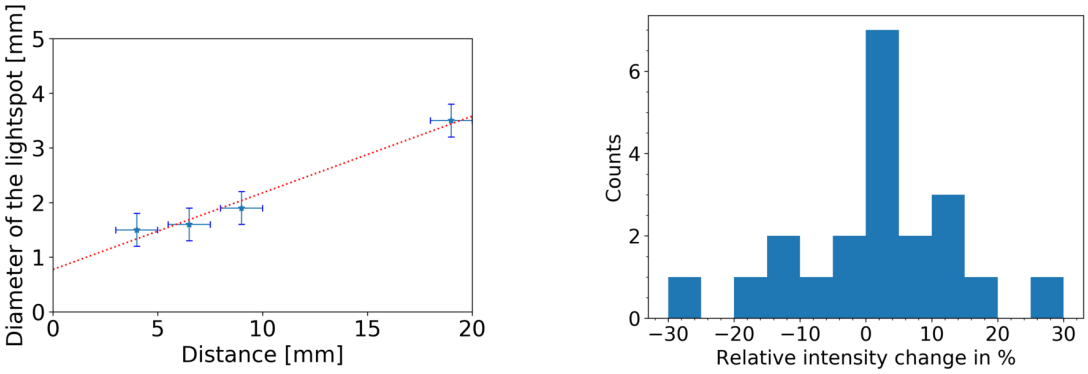


Fig. 4.6: Left: Diameter of the light cone $d_{68\%}$ in dependency of the distance l between mask and light source. Right: Distribution of all relative changes in the intensity of the LED due to a jump observed in the 285 h.

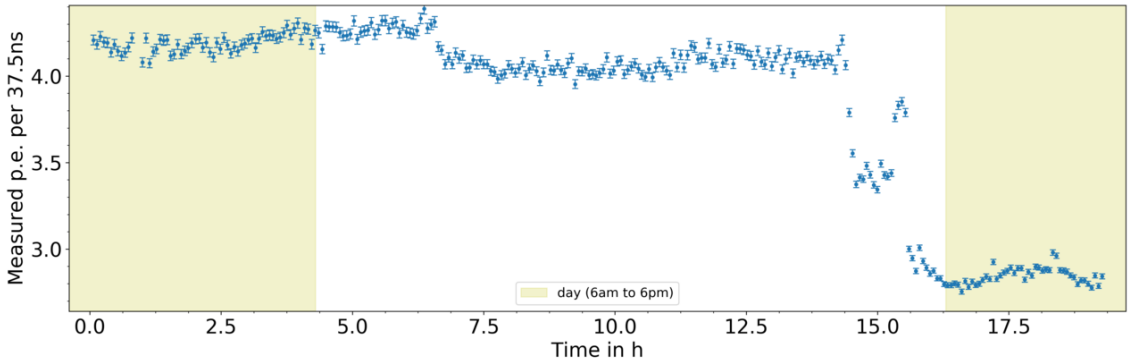


Fig. 4.7: Stability measurement overnight. For visibility, only every fifth data point is shown. The shaded area in yellow markers the time of day between 6 am and 6 pm.

4.3 Functional demonstration with an IceCube-PMT

To test the setup a two dimensional surface scan of an IceCube-PMT[13] was done. Those PMTs were already characterized in [Abb10]. The spatial resolution was 5 mm in this work. The resolution in [Abb10] was 10 mm. The distance of the light source varied between (15 ± 5) mm and (25 ± 5) mm because of the ellipsoidal shape of the PMT. The diameter $d_{\leq 68\,\%}$ of spot were between (2.9 ± 0.7) mm at the largest distance from the photocathode's center and (4.3 ± 0.8) mm at the center. The measurement took 42 h, and the number of collected events per illumination was 10^5. On the left of Figure 4.8, the averaged sensitivity from [Abb10], and on the right, the result from this scan is shown. The positional coordinates measure the distance from the center of the PMT along the curved surface.

[13]Photomultiplier of the type R7081-02 from the company Hamamatsu, Serialnumber: R7081-100-10

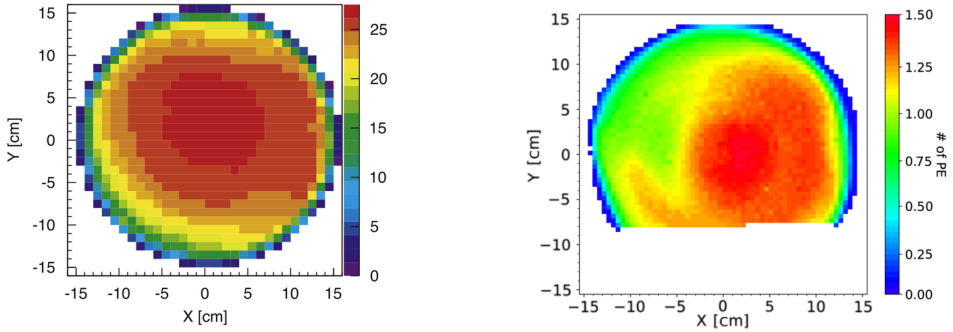


Fig. 4.8: The results of the two dimensional surface scans. The sensitivity is shown as the number of collected photons per light pulse. Left: The average result over 135 different PMTs of the type R7081-02 with a resolution of 10 mm from [Abb10]. Right: The scan was done in this work with a resolution of 5 mm.

The Five Axis Scanner was not able to scan the whole surface of the PMT. For the lower part, ($Y \leq -75mm$), the Scanner would have had to move outside its defined volume. Reference measurements at a fixed point taken sequentially with the actual measurement to record the variations of the light source would increase the measurement time by at least a factor of two. It was decided that the variations during the measurement should be neglected. Nevertheless, the qualitative results from [Abb10] could be reproduced. The authors described a favorable direction due to the alignment of the first dynode. In the scan done in this work, this favorable direction is visible as well. Further, the scan showed that the setup can scan a three-dimensional optical module with a spherical shape with more than 2500 different points over 40 hours automatically.

5. WOM Characterizations

The following chapter describes the method to characterize the light guides. Additional to the WOMs produced in this work (see Chapter 3), three light guides coated in [BK17][1] were characterized as well. At the beginning of this chapter, the methods to handle the raw data from the data acquisition unit are described. After the explanation of the efficiency determination, the systematic uncertainties are discussed.

5.1 Raw data analysis

The data acquisition unit measures the voltage output of the PMTs as a function of the time and sends it to the computer. The data is encoded as an array of 16-bit integer[2] with a length of the number of collected events times the event length in nanoseconds. The time delay between the signal of the light pulse and the trigger can change from measurement to measurement. The integration range to estimate the charge needs to be found for each dataset. Further, all events in a dataset have an individual offset that needs to be found. In this section, the algorithm used to estimate the integration range of a dataset and the offset are presented.

5.1.1 Waveform integration

The time delay between an occurring signal and a trigger pulse depends on the used PMTs, cable lengths, and time resolution of the investigating optical module. Further, the peak in the waveform can be after pulses. If the pulse gets a higher delay, parts of the charge could be lost. The length of the integration range affects the width of electronic noise in B and, therefore, the width of the pedestal (see Section 2.2). It is important to have the integration range at a constant length to guarantee comparability. An algorithm is needed to find the integration range automatically during the measurement. For this, the average of all waveforms has been taken [Sal17]. A waveform with a single photoelectron peak and the average of all waveforms can be seen in Figure 5.1. The position of the maximum has been estimated to find the optimal integration range. It is assumed that the peak in the average is only from the signal. This average corresponds to the time distribution of the signal. The afterpulses produce the asymmetry on the right of the distribution. To have a

[1] PMMA rods with a diameter of 20 mm and a length of 500 mm produced by Grünberg Kunstoffe.
[2] To convert ADQ units into voltage it is divided by the resolution of 2^{16} and multiplied with the measurement range which was in this work ≈ 1 mV.

dynamical measure, the shape of this distribution is assumed to be Gaussian in the first order, and the width σ_S is estimated with the full width of the half maximum (FWHM). The left boundary (left red dashed line) of the integration range is now set to be five times the calculated width σ_S before the maximum.[3] The right boundary is defined with the fixed integration length of 110 ns. The length is set so that it is guaranteed that the afterpulses are included.

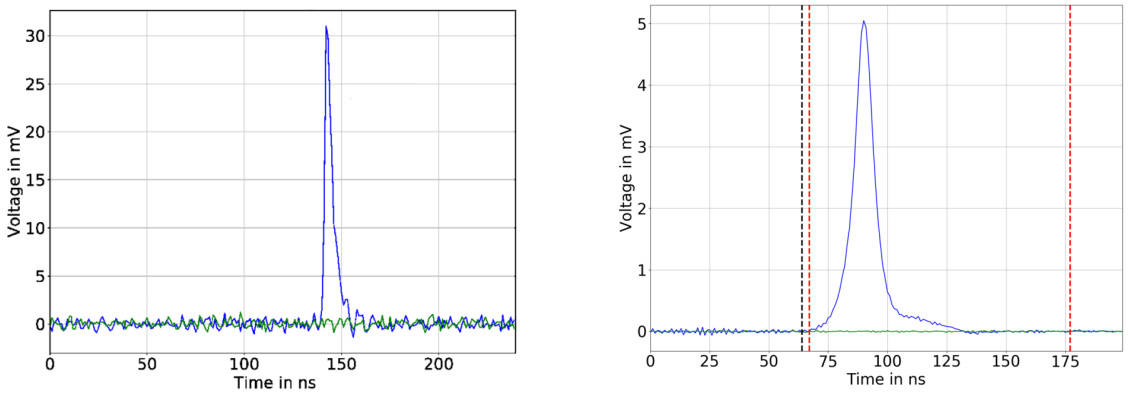


Fig. 5.1: Left: Two example waveforms with a single photoelectron peak (blue) and electronic background (green). Right: The average overall 3×10^5 waveforms. The red dashed lines show the estimated integration boundaries, and the black dashed show the time the ADQ was triggered.

5.1.2 Offset correction

Before integrating the waveforms, the offset of the baseline needs to be estimated for each event. Otherwise, the charge distribution of the estimated T and B are distorted in a nontrivial way. The offset depends on different parameters, such as the current electrical potential on the measuring channel of the ADQ, electromagnetic crosstalk of the cables with ambient electronic devices, and low-frequency noise on the main voltage. The left of Figure 5.2 shows the estimated offset of the waveforms in a dataset in dependence on their measured time. The standard method would be to average the first nanoseconds of the waveform in which no signal is expected. This works as long as the noise of dark pulses is negligible. A dark pulse in the averaged region leads to an overestimation of the offset. In consequence, the integrated charge will be underestimated, and the estimated distributions of T and B are distorted. This artifact of events with underestimated charge can be seen in the right of Figure 5.2. The same dataset with the correct offset was shown in Figure 2.5. Instead, the sum of the minimal value U_{min} of the waveform and the expected Peak-To-Peak value U_{PP} of the electronic noise is taken as a threshold to exclude the photoelectron peaks

[3]Empirical found value.

from the average. Figure 5.3 shows two waveforms from the datasets T and B. The samples between 5 ns to 10 ns left and right of the discarded peaks are removed, as well as the samples to the right from the integration range found in Section 5.1.1.[4] In the end, the offset is calculated from the left samples (shaded samples) of the waveforms.

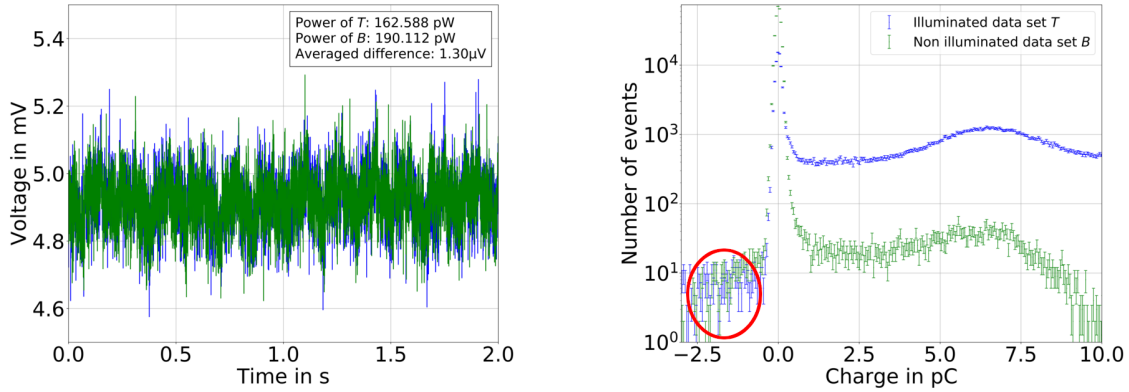


Fig. 5.2: Left: The estimated offset of a dataset over time. Right: Charge histogram distorted by dark pulses of the distributions T (blue) and B (green). Red circle: Underestimated charges. Compare the same datasets with correct offsets in Figure 2.5.

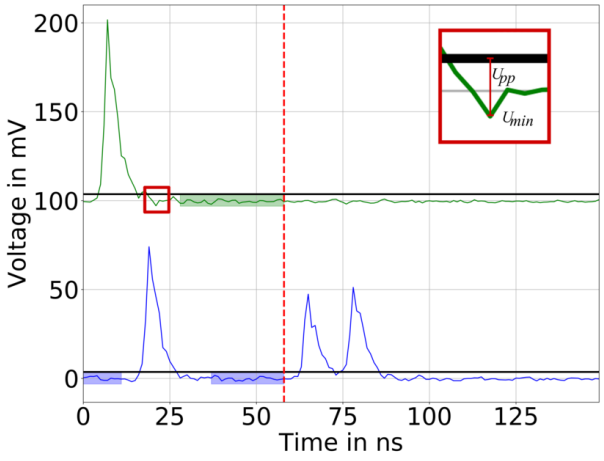


Fig. 5.3: Two waveforms out of the dataset T (blue) and B (green) with multiple pulses. For visibility, 100 mV were added to the waveform of B. The samples with transparent backgrounds are those that are taken to calculate the offset. The samples above the threshold $U_{\text{min}} + U_{\text{PP}}$ (black line), 5 ns/10 ns left/right from a pulse or after the left boundary of the integration range (red dashed line) are discarded.

[4]The assumed Peak-To-Peak value U_{PP} of the electronic noise and the cuts left and right from the peaks are empirical found values which are chosen such that the power and the averaged difference between the offsets of the datasets T and B (see left of Figure 5.2) are minimized. The events of both datasets are collected sequentially, and their offset curve should, therefore, overlap in the optimum.

5.2 Measurement of the WOMs' efficiency

In this section, the characterization of the WOM will be introduced. After the calculation of the correction factor, κ_{corr} from Section 2.1.1, it will be explained how the radial symmetry of a WOM is investigated from the one-sided efficiency ε^{PMT} afterward, the overall efficiency will be estimated.

5.2.1 Measurement procedure

To estimate the one-sided positional dependency of the capture efficiency, $\varepsilon^{\text{PMT}}(z, \Phi)$ the measured light intensity on the ends has to be compared with the actual light intensity of the light. The PMTs are illuminated directly to determine the intensity of the light source. The WOMs were illuminated on different points (z, Φ) perpendicular to the light source. The captured light was detected with the two PMTs on both sides. It was assumed that the light on the WOM's ends is homogeneously distributed. Further, the variation of 1 mm of the centering of the WOM is seen to be negligible. The wavelength dependency of the spatial sensitivity is neglected as well. The light source is assumed to vary linearly over the measurement. Therefore reference measurements has been taken before $\langle \mu^{\text{PMT}} \rangle_{ref,1}$ and after $\langle \mu^{\text{PMT}} \rangle_{ref,2}$ each WOM illumination. For this, the photocathodes were illuminated at multiple points. In the case of tubes, these points $\mu^{\text{PMT}}_{ref,i}(\Phi)$ are aligned in a circle with the WOM's diameter minus its wall thickness. For a rod, the points $\mu^{\text{PMT}}_{ref,i}(r)$ formed a line from the center of the cathode to the boundary of a circle with the diameter of the rod without specific direction. After the first reference measurements, the WOM was mounted between the PMTs. The WOM was illuminated $\mu^{\text{PMT}}(z, \Phi, t)$ at different azimuth angles $\Phi \in [-90\,°, +90\,°]$ and different axial coordinates $z \in [-140\text{ mm}, +140\text{ mm}]$. For all illuminations, the time stamps t have been recorded as well. After the scan, a second reference measurement has been taken to take the linear drift of the light source into account.

5.2.2 Correction factor to involve the spectral sensitivity

To calculate the efficiency, the light intensity captured by the WOM has to be divided by a reference measurement. That measurement must be taken with the same light source. The wavelength-dependent sensitivity Q_{PMT} of the PMT (see Figure 5.4) is given in a resulting current per illuminated power. The spectra I_{LED} and I_{WLS} of the light source and the WLS paint are given as a power per nm. Both spectra are normalized by their integral to one. The absolute units are canceled out in the ratio κ_{corr} (see Equation 2.2b):

$$\kappa_{cor} = \frac{\int_0^\infty d\lambda Q_{\mathrm{PMT}}(\lambda) I_{\mathrm{LED}}(\lambda)}{\int_0^\infty d\lambda Q_{\mathrm{PMT}}(\lambda) I_{\mathrm{WLS}}(\lambda)} = 0.98 \pm 0.02 \tag{5.1}$$

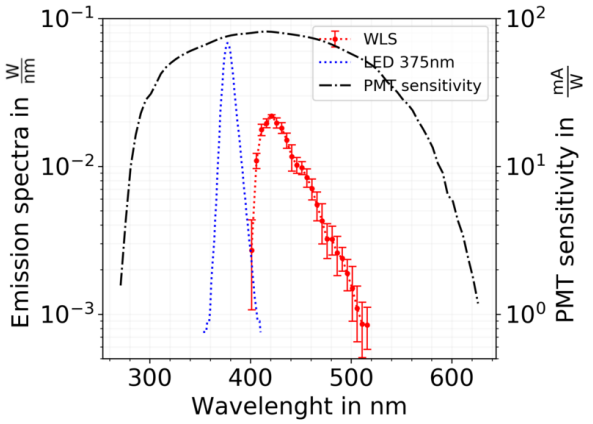


Fig. 5.4: Red and blue line, left axis: The emission spectra of the 375 nm LED XSL-375-3E given by the manufacturer 'Roithner LaserTECHNIK GmbH' and the wavelength shifting paint measured by [Heb14]. Black line, right axis: Sensitivity spectrum of the photomultiplier tube XP2020 given by its manufacturer Philips.

5.2.3 Determination of the position-dependent capture efficiencies

The actual reference values $\langle \mu_{ref,i}^{\mathrm{PMT}} \rangle_\Phi$ for the tubes are the average of all single measurements $\mu_{ref,i}^{\mathrm{PMT}}(\Phi)$.

$$\langle \mu_{ref,i}^{\mathrm{PMT}} \rangle_\Phi = \frac{1}{N_\Phi} \sum_{\Phi \in [0,360]} \mu_{ref,i}^{\mathrm{PMT}}(\Phi) \tag{5.2}$$

N is the number of all single illuminations at the PMT, and Φ is the polar angle of a circle. In the case of the rods, the ten points form a line from the center of the cathode to the boundary of a circle with the rod's diameter. It was assumed that the sensitivity of the PMT is radial symmetrical around its center, and the average of the single points has been calculated by weighting them with the distance r to the center of the cathode. The single point $\mu_{ref,i}^{\mathrm{PMT}}(r)$ are seen as representative for a ring of the width $\Delta = \frac{r_{\mathrm{WOM}}}{N-1}$ with r_{WOM} as the WOM's radius. The weighing factors $w(r)$ are the ratio of the ring's area $A(r)$ with the overall covered area by the WOM $A_{\mathrm{WOM}} = \pi r_{\mathrm{WOM}}^2$.

$$\langle \mu_{ref,i}^{\mathrm{PMT}} \rangle_A = \frac{1}{A_{\mathrm{WOM}}} \sum_r A(r) \mu_{ref,i}^{\mathrm{PMT}}(r) \tag{5.3}$$

All illuminations are done at a distance such that the diameter of the illuminating spot has been the same as the wall thickness of the tube (see Section 4.2.1). In the case of rods, the diameter of the spot was chosen such that it is the same as the width Δ. Figure 5.5 shows a usual reference measurement $\mu_{ref,i}^{\mathrm{PMT}}(\Phi)$ of one tube.

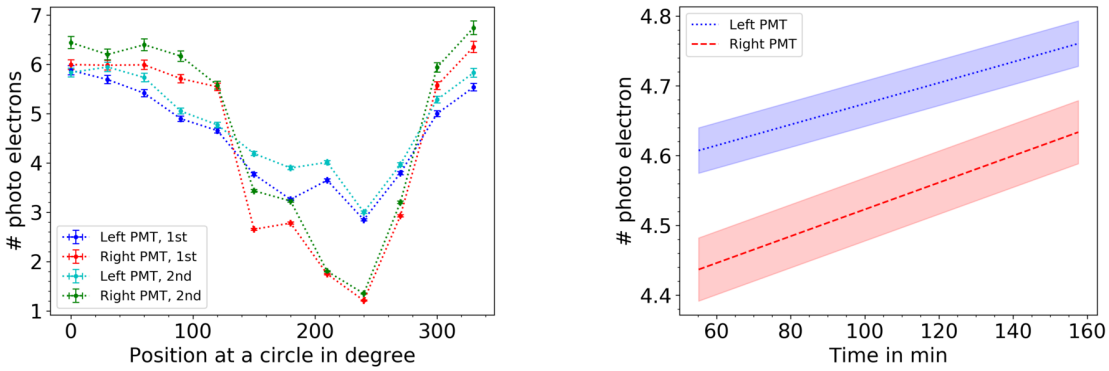


Fig. 5.5: Left: Reference measurements before and after the illumination of the WOM P45/5-6/3 for both PMTs. Right: Linear shift of the reference values for both PMTs.

The measurement of the reference values $\mu_{ref,i}^{\mathrm{PMT}}(\Phi)$ is taken in around 20 minutes. It is assumed that the variation of the light source during this time period is negligible. The variations in the measured photoelectrons are due to the inhomogeneity of the sensitivity of the photocathode. As long as the light outcome of the WOM can be seen as homogeneous, the average Equation 5.2 takes this inhomogeneity into account. The different shapes between the reference measurements before and after the WOM illumination are a consequence of the unavoidable variation of the alignment of the PMT with respect to the Scanner's coordinate system. This error will be discussed in Section 5.4.1.
After both reference measurements, a linear regression was done to estimate the light intensity of the light source at the moment t of illumination of $\mu(z, \Phi, t)$. Figure 5.5 shows the assumed linear drift of the reference values during the WOMs illumination of both PMTs. The errors given here are statistical errors given by the measurement method (see Section 2.2.1) and calculated with Gaussian error propagation. To get the spatial depended one-sided capture efficiencies, the measured light $\mu^{\mathrm{PMT}}(z, \Phi, t)$ from the WOMs illumination is divided by its corresponding time-dependent reference value $\mu_{ref}^{\mathrm{PMT}}(t)$. The one-sided capture efficiency $\varepsilon^{\mathrm{PMT}}(z, \Phi)$ follows now from Equation 2.2c:

$$\varepsilon^{\mathrm{PMT}}(z, \Phi) = \frac{\mu^{\mathrm{PMT}}(z, \Phi, t)}{\mu_{ref}^{\mathrm{PMT}}(t)} * \kappa_{corr} \tag{5.4}$$

Figures 5.6 show usual results of the surface scan of the WOM P45/5-6/3.

5.2.4 Radial symmetry

Due to its radial symmetry, the WOM's capture efficiency is assumed to be equal in all directions. Nevertheless, it is possible that the WOM has a favorable direction due

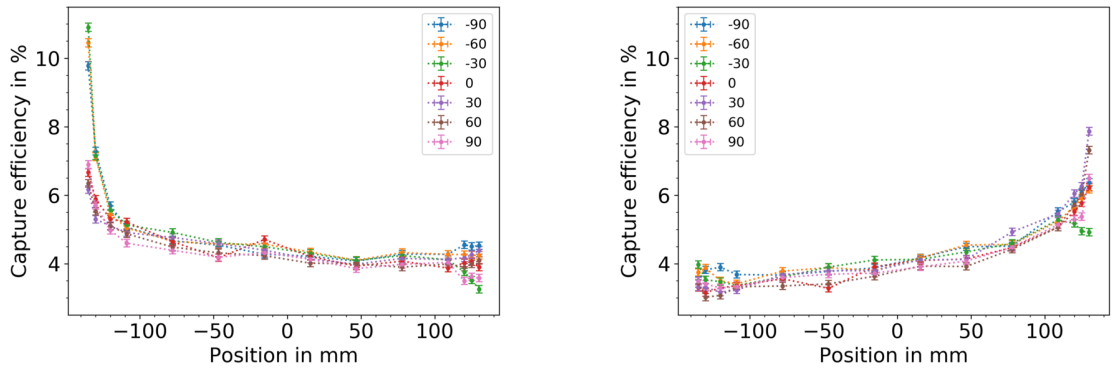


Fig. 5.6: Results of the WOM P45/5-6/3. Left/right figure: results of the left/right PMT. The one-sided capture efficiency $\varepsilon^{PMT}(z, \Phi)$ is plotted against the position at the axial axis on the WOM in mm for different polar angles of the surface. The illumination is always done perpendicular at a distance of (20 ± 5) mm corresponding to a spot of (3.6 ± 0.7) mm in diameter (see Section 4.2.1).

to production errors. In Figure 5.7, the estimated one-sided efficiencies $\varepsilon^{\mathrm{PMT}}(z, \Phi)$ are plotted against the azimuth angle Φ for different distances d to the corresponding PMTs.

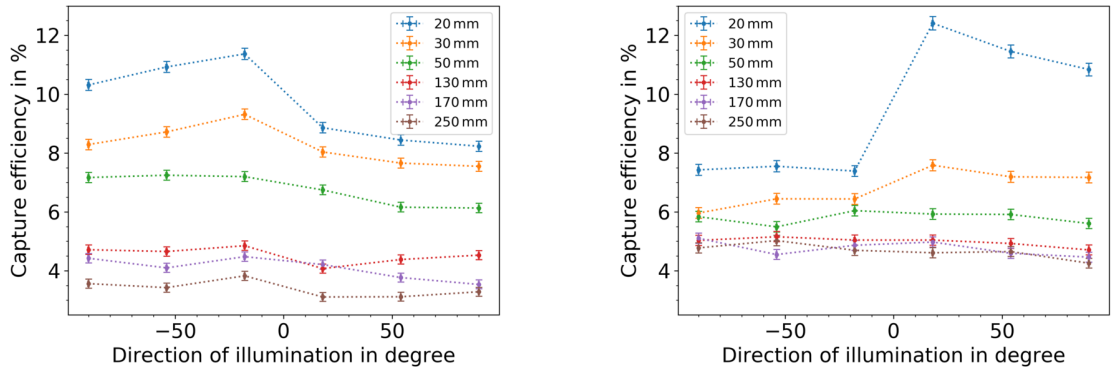


Fig. 5.7: The one-sided capture efficiency $\varepsilon^{PMT}(z, \Phi)$ of the WOM P45/3/4 is shown as a function of the azimuth angle Φ for different distances to the corresponding PMT d.

Figure 5.7 shows that at distances up to 50 mm to the left PMT, more light was detected in the range between $-90°$ and $0°$. The right plot of Figure 5.7 shows the same behavior but with reversed angular dependence. This can be explained by an offset in the azimuth orientation of the mounted LED, which was, therefore, not pointing perpendicular to the WOM's surface. Direct light has illuminated the PMT and led to an overestimation of the WOM's one-sided efficiency. To illuminate the WOM in the range of $0°$ to $90°$, the LED was rotated by $180\,°$, and the offset in the azimuth angle was reversed, which led to an overestimation in the efficiency measured

with the other PMT. This effect was negligible for distances larger than 50 mm from the PMTs.
To quantify the variation of the one-sided capture efficiency $\varepsilon^{\mathrm{PMT}}(z, \Phi)$ over the angle the relative deviation $\delta\varepsilon^{\mathrm{PMT}}(z, \Phi)$ from it's angular average $\langle\varepsilon^{\mathrm{PMT}}\rangle_{\Phi}(z) = \frac{1}{N_{\Phi}} \sum_{\Phi} \varepsilon^{\mathrm{PMT}}(z, \Phi)$ is calculated for all points and for both PMTs. To avoid data that was compromised by direct light from the light source, only measurements with points with a minimal distance of 50 mm to both PMTs were used for analysis.

$$\delta\varepsilon^{\mathrm{PMT}}(z, \Phi) = \frac{\varepsilon^{\mathrm{PMT}}(z, \Phi)}{\langle\varepsilon^{\mathrm{PMT}}\rangle_{\Phi}(z)} - 1 \tag{5.5}$$

Figure 5.8 shows a histogram of all relative deviations estimated for all distances between 50 mm to 250 mm and both PMTs. From this distribution, the standard deviation σ_{Φ} has been calculated. If this variation is based on statistical fluctuations only, the same standard deviation σ_{stat} has to be estimated by using Gaussian error propagation for the average of all $\delta\varepsilon$.

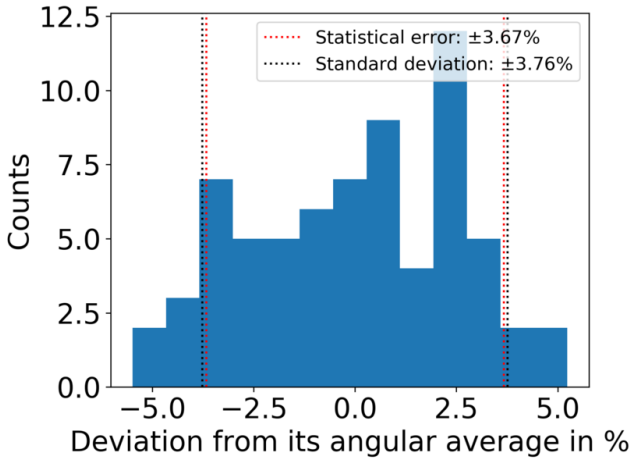


Fig. 5.8: Histogram over the relative deviations of all measured points. Red dashed line: The statistical error σ_{stat} estimated from the error of the single measurements. Black dashed line: The standard deviation σ_{Φ} of this distribution.

In the case the statistic uncertainty of the average σ_{stat} is smaller than the actual variation σ_{Φ}, it can not be explained with statistical fluctuation only, and a deviation from symmetry is possible. There were three rods, P20/1, P20/2, and P20/3, and one tube, P45/4-5/1, that were treated as unsymmetrical (see the left of Figure 5.9). The rods P20/1, P20/2, and P20/3 had a variance σ_{Φ} between 12 % and 14 % while the statistic uncertainties were around 3 %. They were coated in the work [BK17] at once and arranged in a triangle with a distance of $\leq$ 15 mm to each other. In the right of Figure 5.9, the one-sided capture efficiency $\varepsilon^{\mathrm{PMT}}(z, \Phi)$ of rod P20/2 is shown as a function of the direction of illumination Φ for different distances to the detecting PMT. Between the distances of 20 mm and 120 mm a favorable direction can be

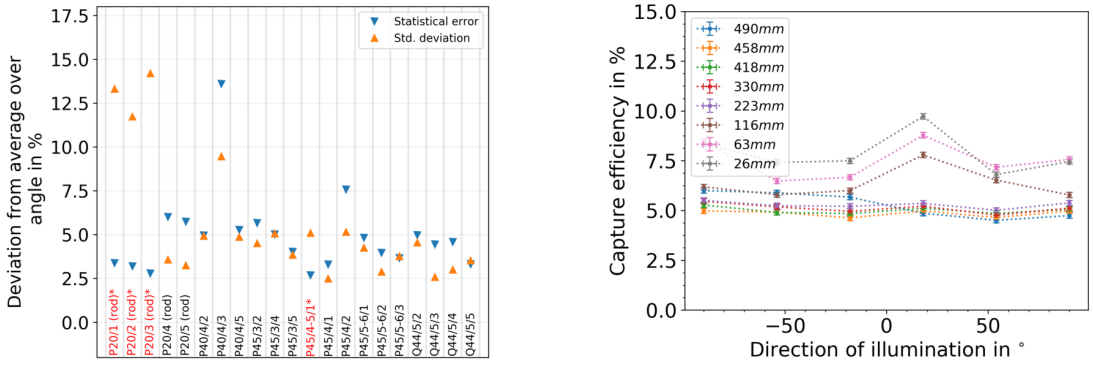


Fig. 5.9: Left: The deviation around it's angular average σ_Φ and the statistic uncertainty of the average σ_{stat} of all measured WOMs. The WOMs treated as unsymmetrical are colored red. Right: The one-sided capture efficiency $\delta\varepsilon^{\text{PMT}}(z, \Phi)$ of rod P20/2 is shown as a function of the angle of illumination for different distances to the detecting PMT.

observed in the range of $-20\,°$ and $50\,°$.[5] The rods P20/1 and P20/3 have shown the same behavior. It is possible that the meniscus of the paint was inhomogeneous, which made the paint on the side facing the others thicker. The presence of the other WOMs had disturbed the paint to drain off on this side [BK19]. In conclusion, it is necessary that the distance between WOMs coated at once is large enough that the meniscus of the paint is symmetric around the WOM during its withdrawal. Furthermore, it is possible that the paint thickness was in the same order as its absorption length. The absorption coefficient a_{ABS} was less than 100 % and still varied over the paint's thickness. Figure 5.10 shows the tube P45/4-5/1 under UV light. It had multiple uncoated regions, which can be seen as shadows. It is possible that the cleaning process was insufficient for this tube. Nevertheless, the asymmetry had a significance of 2σ, and the variance is still compatible with the averaged variance of

$$\langle \sigma_\Phi \rangle = (4.2 \pm 1.6)\ \%$$

from all WOMs treated as symmetric. The averaged statistic uncertainty was $\langle \sigma_{\text{stat}} \rangle = 5.4\ \%$. The maximal observed variance was $\sigma_\Phi^{\text{max}} = 9.5\ \%$ and the minimal was $\sigma_\Phi^{\text{min}} = 2.5\ \%$.

Even if favorable sides have not been observed in the case of the tubes, surfaces with distances less than 15 mm to the WOM's surface should be avoided. To tighten the tubes, it is therefore recommended that only one threaded rod should be passed through the tube instead of the option used in this work (see Chapter 3) to avoid possible inhomogeneities.

[5] It can be excluded that the signal is produced by the direct PMT illumination described above because it was also observed in the other PMT and decreased after 50 °.

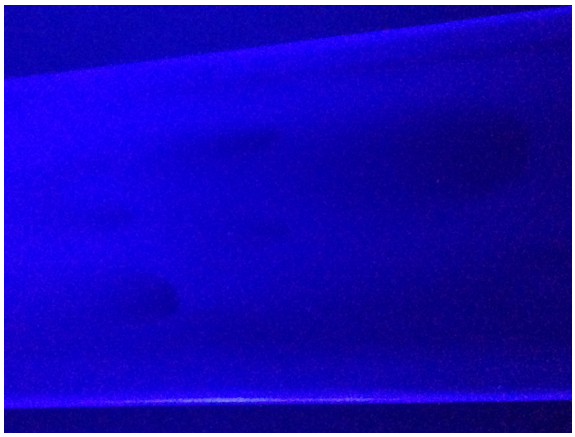


Fig. 5.10: WOM P45/4-5/1 illuminated with UV light. The uncoated regions are visible as shadowed areas.

5.3 Geometry

In the measurements of the capture efficiency, it was assumed that the light is distributed homogeneously over the WOM's ends. To verify this assumption, the relative light distributions at both ends were measured. Masks were prepared from black tape[6] and aluminum foil to cover the PMTs with except a slit of 3 mm (see. bottom of Figure 5.11).

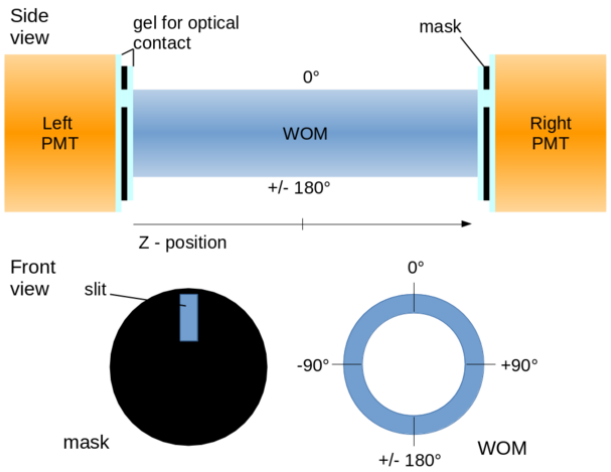


Fig. 5.11: Schematic of the setup to measure the light distribution on the ends of a cylindrical light guide.

Between WOM and mask, a gel pad, as described in the appendix (see Section A.2), was used to increase the optical contact. Instead of rotating the mask between WOM and PMT, a two dimensional surface scan was done with the coordinates $\Phi \in [-90, -80, ..., 80, 90]$ (see bottom Figure 5.11) and $z \in [-130, -120, ..., 120, 130]$ (see top Figure 5.11). It was assumed that the WOM is radially symmetric. Due to the variation of the light source over time a reference value $\mu_{\text{ref}}^{\text{PMT}}(i_t)$ was measured

[6]Manufacturer: Tesa SE

between each new scanned point $\mu^{\mathrm{PMT}}(z, \Phi, i_t)$, like it was done in the surface scans of the PMTs. The distance of the light source to the WOM was (20 ± 5) mm. The diameter $d_{68\,\%}$ of the illuminating spot was thus (3.6 ± 0.7) mm. The Five Axis Scanner was only able to scan half of the cylinder. To get a complete scan of the WOM's surface, it was rotated by 180° together with the PMTs, and the masks and the scan were repeated. In the first/second measurement, the slits were at the angular coordinates $\Phi = \pm 90°$. The coordinates for the reference values were $Z = 0$ and $\Phi = \pm 90°$. By repeating the scan, the points with the azimuth angles $0\,°$ and $180\,°$ were measured twice, which guarantees comparability. The whole scan was done three times, and the relative sensitivities $\varepsilon_{rel}^{\mathrm{PMT}}(z, \Phi)$ with respect for the reference point were calculated using Equation 5.7. Figure 5.12 shows the averaged results of the scans in dependency of the angle of illumination Φ for different distances to the corresponding PMTs. The angle Φ is with respect to the position of the slit as described in the bottom figure of 5.11).

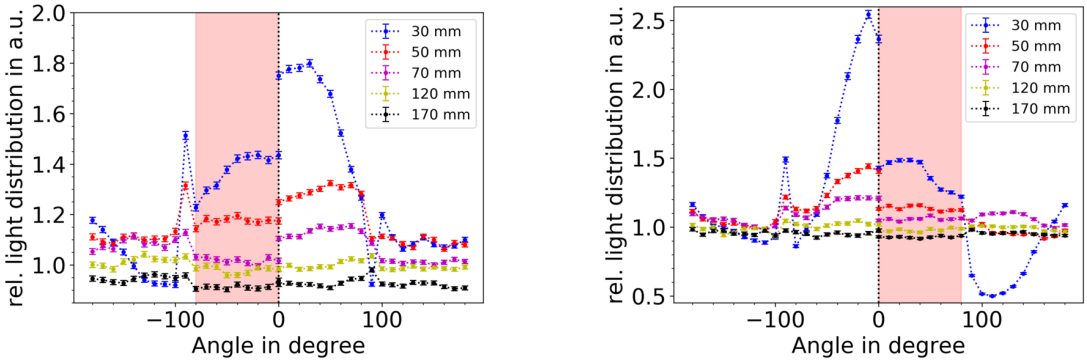


Fig. 5.12: Results of the angular distribution scan. The relative intensities $\varepsilon_{rel}^{\mathrm{PMT}}(z, \Phi)$ in dependence of the azimuth angle between the slit and the point of illumination Φ. Different distances d to the PMTs are shown.

In Figure 5.12 the estimated relative intensities are shown for different distances d to the PMTs in dependence of the azimuth angle Φ. The data can be divided into quadrants with the first (I) from $-180\,°$ to $-90\,°$ and the second (II) from $-90\,°$ to $0\,°$ and so on. In the quadrant $\mathrm{I}_{\mathrm{left}}$ and $\mathrm{III}_{\mathrm{left}}$ from the left and $\mathrm{II}_{\mathrm{right}}$ and $\mathrm{IV}_{\mathrm{right}}$ from the right PMT the measurements are compromised by direct light from the LED for the same reason as explained in Section 5.2.4.
To quantify the inhomogeneity the p-value was calculated to test the hypothesis that there is no dependency on the angle Φ and all values are therefore statistical distributed around the angular average of the relative intensities $\langle \varepsilon_{rel}^{\mathrm{PMT}} \rangle_{\Phi}(z)$. The $\chi^2(z)$ was calculated:

$$\chi^2(z) = \sum_{\Phi} \left(\frac{\varepsilon_{rel}^{\mathrm{PMT}}(z, \Phi) - \langle \varepsilon_{rel}^{\mathrm{PMT}} \rangle_{\Phi}(z)}{\Delta \varepsilon_{rel}^{\mathrm{PMT}}(z, \Phi)} \right)^2 \tag{5.6}$$

From this the p-value was estimated with $p = 1 - F(\chi^2, k)$ whereby F is the cumulative distribution function and k the degrees of freedom. For analysis the quadrants $\mathrm{II}_{\mathrm{left}}$ and $\mathrm{III}_{\mathrm{right}}$ are used instead of $\mathrm{IV}_{\mathrm{left}}$ and $\mathrm{I}_{\mathrm{right}}$ because the first two shows a stronger angle dependence. In Figure 5.13 the p-values are shown in dependency of the distance d to the corresponding PMTs. It can be seen that the model of a homogeneous light distribution can be falsified up to a distance of

$$(50 \pm 10)\ \mathrm{mm}$$

from the PMT. It can be concluded that the light distribution is homogeneous if the point of illumination is at least 50 mm away from the WOM's ends.

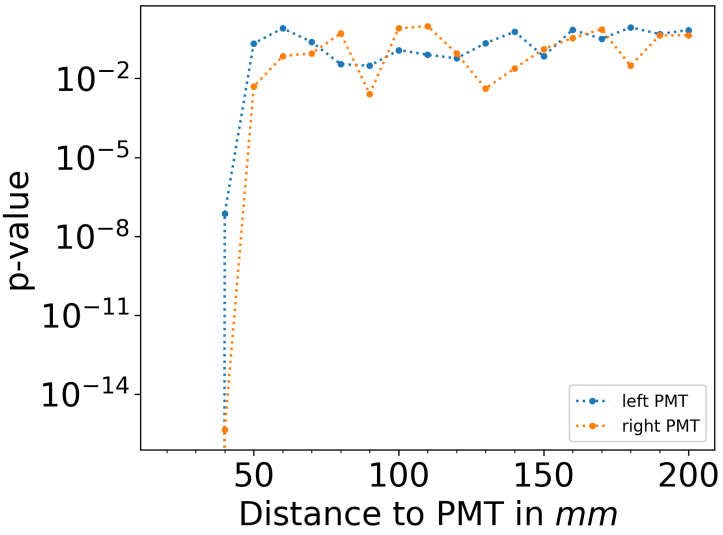


Fig. 5.13: P-value of the hypothesis of an angular independent light distribution at the ends of the WOM in dependence of the axial coordinate z.

5.4 Systematic uncertainties

In this section, the methods to investigate the systematic uncertainties are described. The influence of the inhomogeneity of the photocathode of the used PMTs is estimated as well as their nonlinearity. The last uncertainty occurred with the nonlinear random jumps of the intensity of the light source described in Section 4.2.2.

5.4.1 Photocathodes' spatial variation

For each reference measurement, the PMT had to be rotated by 90 ° to allow the Scanner to point at the photocathode. The illumination had to be done over the same area the WOM had contact with. It was assumed that the variation of the sensitivity has no significant effect on the measurement. In the efficiency measurements done

in Section 5.2.3, it was assumed that the alignment of the WOM and the reference scan with respect to the PMT did not vary significantly and had no effect on the measurement. In the reference measurements, it was observed that the sensitivity of the photocathode varied over a factor of six (see Figure 5.5). This strong positional dependence of the cathode's sensitivity motivated to verify the above assumption afterward.
The relative sensitivity of the photocathodes were measured by scanning the cathodes with the light source and estimating the variation over the surface $\epsilon_{rel}^{\mathrm{PMT}}(x, y)$. With x and y as the positions on the PMT's photocathode. A Monte Carlo simulation was done with a random variation of the center of the WOM's center position. The variation of the relative averaged sensitivity of this area on the alignment was estimated. The scan was done for both PMTs at a distance of (5 ± 2) mm which results in a spot with a diameter $d_{68\leq\%}$ of (1.5 ± 0.3) mm and a spatial resolution of 1 mm. Due to the variation of the light source between each illumination, the center of the photocathode was illuminated as a reference value. Figure 5.14 shows this reference values $\mu_{\mathrm{ref}}^{\mathrm{PMT}}(i_t)$ over time from a two dimensional surface scan of the left PMT.

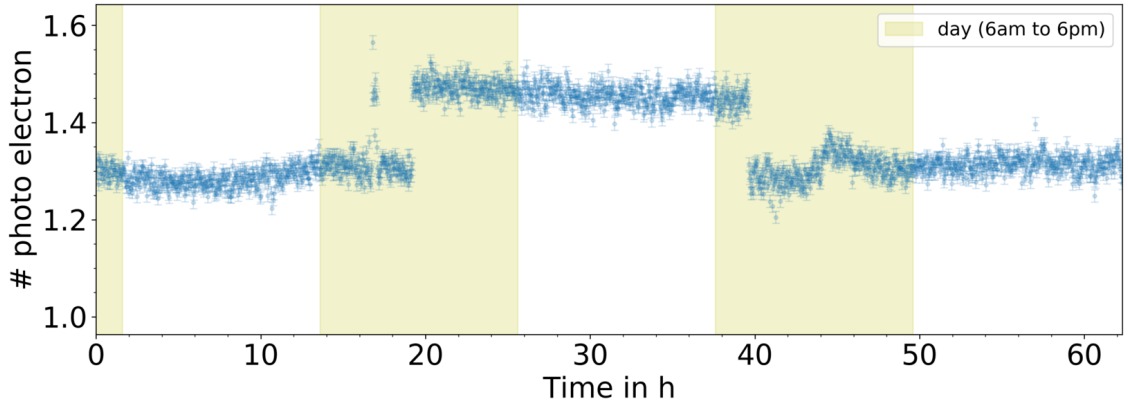


Fig. 5.14: The reference values taken during the two dimensional surface scan of the left PMT plotted against real time. For the visibility the data points are transparent.

The values from the two dimensional scan were normalized by the average of two sequently measured reference values. The absolute spatial measurements $\mu^{\mathrm{PMT}}(x, y, i_t)$ are numbered with the time steps i_t. The reference values $\mu_{\mathrm{ref}}^{\mathrm{PMT}}(i_t)$ have half-integer time steps.

$$\epsilon_{\mathrm{rel}}^{\mathrm{PMT}}(x, y) = \frac{2\mu^{\mathrm{PMT}}(x, y, i_t)}{\mu_{\mathrm{ref}}^{\mathrm{PMT}}(i_t - \frac{1}{2}) + \mu_{\mathrm{ref}}^{\mathrm{PMT}}(i_t + \frac{1}{2})} \tag{5.7}$$

The left PMT was scanned three times, and the right one was scanned two times. Each scan took ≈ 62 h. The resulting relative sensitivities were averaged. The figures

5.15 show the estimated positional sensitivities normalized with respect to the center of the photocathode.

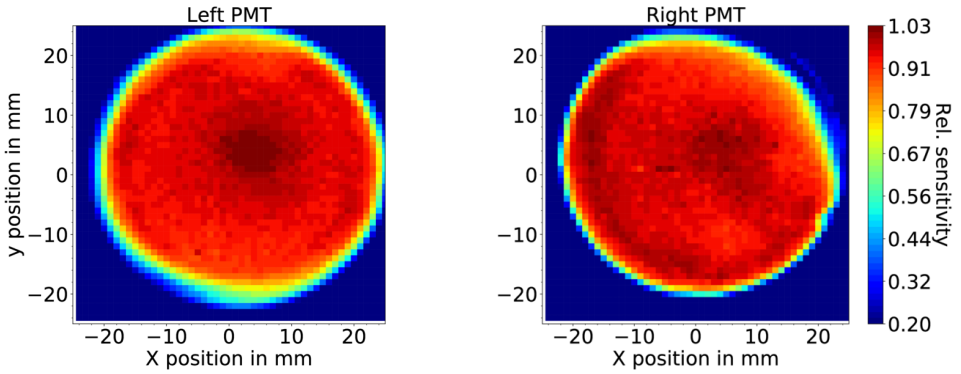


Fig. 5.15: Resulting relative sensitivities from all two dimensional surface scan for the left and the right PMT.

Figure 5.16 shows the profile of the estimated relative sensitivities. The shaded region shows the WOM's covering area, which contributes to the averaged sensitivity. It can be seen that the WOM with a radius R and a wall thickness w has contact with the photocathode where the PMT loses its sensitivity significantly down to 50% with respect to its center. To estimate the error of the lack of precise alignment between the WOM and the reference measurement, the variation in centering of the contacting area was simulated in a Monte Carlo simulation. It was assumed that the variation of the center is normally distributed in x and y direction with a standard deviation of 1 mm.[7]

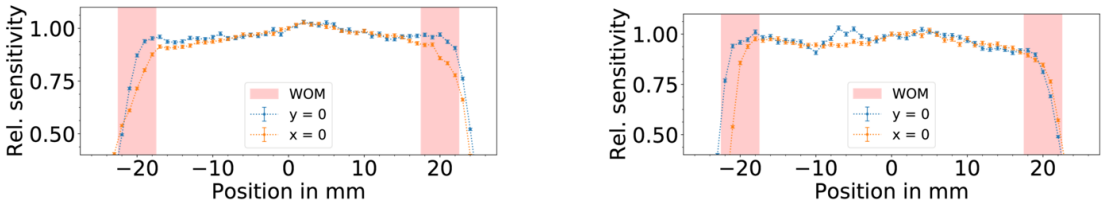


Fig. 5.16: Profile of the relative sensitivities through the x and y axis crossing the center of the photocathode. Shaded: covered area of a WOM tube with an diameter of 45 mm and a wall thickness of 5 mm.

From the center the average of all values covered by the WOM's ends with a wall thickness w and a radius R the relative sensitivity was calculated:

[7]To estimate this value the light source was pointed at a reference point on the PMT mounts to verify the alignment of the Scanner's coordinate system before each reference measurement was done. No deviation larger than ±1 mm was observed.

$$\langle \epsilon^{rel} \rangle (x_0, y_0) = \sum_{r,\Phi} \epsilon_{rel}^{\mathrm{PMT}}(r, \Phi) \tag{5.8}$$

Whereby x_0 and y_0 are the normal distributed coordinates of the WOM's center, $r \in [R-w, R]$ and $\Phi \in [0, 360)$ the polar coordinates with (x_0, y_0) as origin, $\epsilon_{\mathrm{rel}}^{\mathrm{PMT}}(r, \Phi)$ the relative sensitivity estimated from the scan above and $\langle \epsilon_{\mathrm{rel}}^{\mathrm{PMT}} \rangle (x_0, y_0)$ the averaged sensitivity over the covered area with respect to the sensitivity to the center of the photocathode. In Figure 5.17 the covered points by the WOM are marked if the center lies in the expectation value $(0, 0)$ on the left PMT and the outcome of a Monte Carlo simulation x_0 and y_0 being normal distributed with a standard deviation of 1 mm.

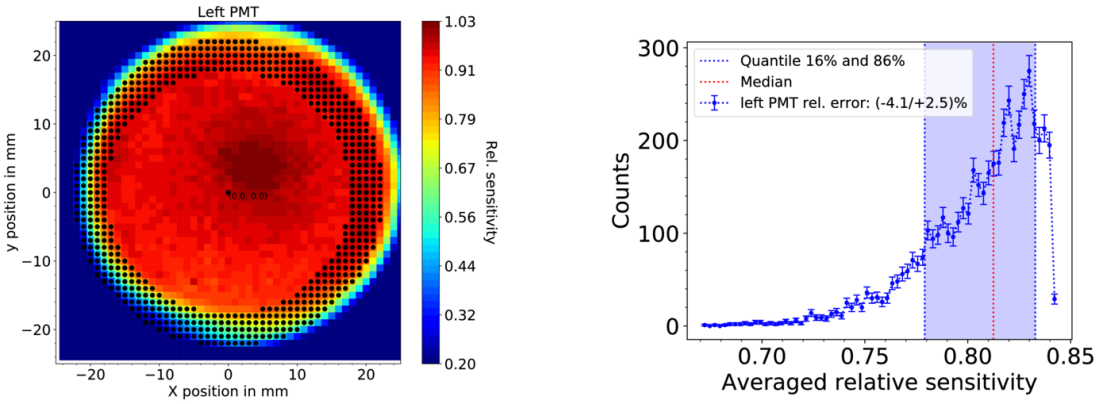


Fig. 5.17: Left: all averaged points of the relative sensitivity $\epsilon_{\mathrm{rel}}^{\mathrm{PMT}}(r, \Phi)$ to estimate $\langle \epsilon_{\mathrm{rel}}^{\mathrm{PMT}} \rangle (0, 0)$ for the left PMT and a WOM with a radius $r = {}^{45\,\mathrm{mm}}/_{2}$ and a wall thickness $w = 5$ mm. Right: histogram of the outcome of 5000 trials of $\langle \epsilon_{\mathrm{rel}}^{\mathrm{PMT}} \rangle (x_0, y_0)$ for x_0 and y_0 being normal distributed.

To estimate the systematic uncertainty, the quantiles Q_{16} and Q_{86} at 16% and 86% of the distributions are assumed to mark the error bands [Lis16]. As long as the average is only justified, the expectation value for normally distributed errors [Tay88] the median Q_{50} is assumed to be the most likely value. In the end, the relative error is estimated by calculating:

$$\Delta_- = \frac{Q_{16}}{Q_{50}} - 1 \tag{5.9}$$

$$\Delta_+ = 1 - \frac{Q_{86}}{Q_{50}} \tag{5.10}$$

This is done for all needed geometries.
As it is investigated in Section 5.4.1 the uncertainty of 1 mm in the alignment of the WOM have a significant effect on the measurement due to the inhomogeneity of the photocathode. The estimated uncertainties for the different shapes of the

WOMs are listed in Table A.1 with the photocathodes of the used PMT. The largest uncertainties for the PMT are $^{+5.3}_{-6.2}$ %. The uncertainties are larger with smaller wall thickness because the average of a smaller area fluctuates more. The assumption that the spatial sensitivity has no significant influence on the reproducibility of the measurement was wrong. The error bands estimated here are used as systematic uncertainties for the reference measurement.
The variation of the 20 mm rods were less than $\pm 0.2\,\%$ because they had contact only with the part of the photocathode with variations $\leq 7\,\%$. The photocathodes with a diameter of 50 mm were therefore too small for WOMs with a diameter of 45 mm. The ratio between the diameter of the WOM rod and the diameter of the photocathodes is 1.6. It is recommended to have photocathodes larger than 1.6 times the WOM's diameter to minimize this effect. Nevertheless, the spatial sensitivity of a photocathode can vary about 30 % being only one-third of the photocathode's radius away from its center (see Section 4.3). An investigation of the spatial sensitivity is recommended in general for WOM characterizations with aimed relative uncertainties less than 6 %.

5.4.2 PMT linearity

For the measurement of the capture efficiency, $\varepsilon_{\text{eff}}(z, \Phi)$ the measured light intensity $\mu(z, \Phi)$ is compared with the reference measurement μ_{ref} which differ by order of magnitude. It is assumed that the photomultipliers behave linearly over this range. To verify the linearity, the PMTs are illuminated with two independent LEDs ($405nm$). Three single measurements were done. For the measurement of μ_{LED1} and μ_{LED2}, the LEDs were switched on one after another. In the measurement of $\mu_{\text{LED1/2}}$, both LEDs have illuminated the PMTs. The deviation of the PMT from linearity can be determined by comparing the sum $\mu_\Sigma = \mu_{\text{LED1}} + \mu_{\text{LED2}}$ with $\mu_{\text{LED1/2}}$) in the range of $min(\mu_{\text{LED1}}, \mu_{\text{LED2}})$ to $\mu_{\text{LED1/2}}$). Both LEDs were embedded in cylindrical acrylic glass connected to an acrylic fiber with a diameter of ≈ 0.5 mm and 10 cm length (see fig. 4.3). The LEDs were covered with black tape to avoid undirected light and to point into the upper hole of an integrating sphere to simulate one light source. The PMTs have had a distance of (35 ± 1) cm to the holes of the sphere. The setup is shown in Figure 5.18. Both LEDs were pulsed with a pulse generator[8]. The width of the pulses was fixed at 37.5 ns. To vary the light intensity, the amplitude of the pulses were varied between 2.69 V and 2.83 V. The Voltages were chosen such that the intensities of both LEDs differed less than 50%. The accuracy was limited by the step width of 0.01 V.

To cover the full range of interest at each measurement of μ^{i+1}_{LED1} and μ^{i+1}_{LED2} the

[8]Stanford Research Systems Model DG645 SN:005665

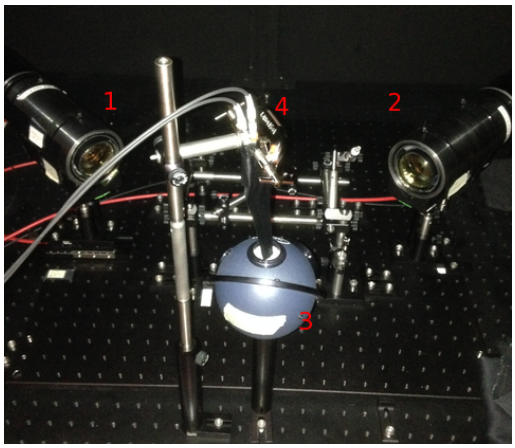


Fig. 5.18: Setup of the linearity measurement. **1**: Left PMT (XP2020, SN23890) **2**: Right PMT (XP2020, SN23998) **3**: integrating sphere made with Teflon and coated with Spectralon from Labsphere. The sphere has three holes. **4**: Two 405 nm LEDs from Roithner Lasertechnik GmbH covert with black tape and pointing into the upper hole of the integrating sphere.

single LEDs had an intensity as they had together in sum before $\mu_\Sigma^i = \mu_{\mathrm{LED1}}^{i+1} = \mu_{\mathrm{LED2}}^{i+1}$. The left of Figure 5.19 shows the difference between the sum μ_Σ and the combined intensity $\mu_{\mathrm{LED1/2}}$ for all measurements up to 8 photo electrons.

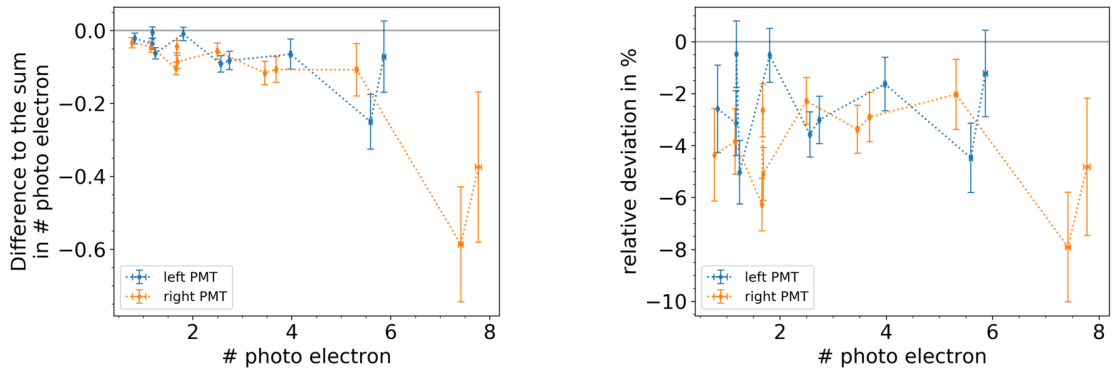


Fig. 5.19: Left/Right: The difference/relative deviation between calculated intensity μ_Σ from single measurements and measured intensity with both LEDs $\mu_{\mathrm{LED1/2}}$.

It can be seen that the PMTs underestimate the number of photo electrons after each doubling of the intensity. To quantify this nonlinearity the average of the relative deviation $\Delta^i = \mu_{\mathrm{LED1/2}}^i / \mu_\Sigma^i$ is calculated weighted with it's errors. The nonlinearity between an intensity and it's double is determined as:

$$\begin{aligned} \langle \delta \rangle_{rel}^{left} &= (-3.7 \pm 0.3)\% \\ \langle \delta \rangle_{rel}^{right} &= (-2.6 \pm 0.4)\% \end{aligned} \tag{5.11}$$

To estimate the maximal error on the capture efficiency one has to compare light intensities differing in $O(1.5)$ orders of magnitude. To quantify the underestimation in the range of the high intensity value μ_{high} with respect to the low intensity value μ_{low} one calculates the number of multiplications with 2. If the PMT has underestimated

the intensity 0.6 starting from 0.3 by factor of $1 - \langle\delta\rangle_{\text{rel}}$ it will underestimate the value for $1.2 = 0.6 * 2$ by $(1 - \langle\delta\rangle_{\text{rel}})^2$ and so on.

$$\Delta = 1 - (1 + \langle\delta\rangle_{rel})^{\log_2(\frac{\mu_{high}}{\mu_{low}})} \tag{5.12}$$

Δ is the expected underestimation of μ_{high} with respect to μ_{low} and therefore the overestimation of the capture efficiency.

The minimal value during the measurement was in the order $\mu_{\text{low}} = 0.3$ photoelectrons and the reference values were in the order of $\mu_{\text{high}} = 8$ photoelectrons. With the values from 5.11 and equation 5.12 the resulting uncertainty due to nonlinearity of the PMTs are shown in Table 5.1. The reference values are therefore underestimated

PMT	$\langle\delta\rangle_{rel}$ in %	Underestimation in %
left	-3.7 ± 0.3	-16
right	-2.6 ± 0.4	-12

Tab. 5.1: caption

by 16 % at the left PMT and 12 % at the right PMT. As a consequence, the one-sided capture efficiencies are overestimated by these factors. Nevertheless, it is possible that the pulse generator which has been used as a power supply for the LEDs is not able to supply with a constant current. The LEDs could therefore have been emitted less light when they were supplied at once. The nonlinearity may be an effect of the trigger generator. As long as this is unknown, these factors are taken as additional systematic uncertainty for the estimated efficiencies. To correct the nonlinearity of the PMTs, they need to be calibrated in the single photoelectron regime over a range of at least one order of magnitude with a calibrated light source.

5.4.3 Nonlinear shift of the light source

As it was discussed Section 4.2.2 the light source did nonlinear fluctuations with sudden changes in intensity up to ±30 %. It is assumed that if such a sudden jump appears during the efficiency measurements. It generates a discrete increase or decrease in the measurements of the one-sided capture efficiency. The signature of this effect will occur in the measurements of both PMTs at the same time, while the WOM is scanned along the axial coordinate z from -150 mm to 150 mm. The one-sided capture efficiencies of all WOMs were examined. A discrete increase of the estimated capture efficiency was observed in the measurement of P45/3/5 and can be seen in Figure 5.20. By comparison of the different efficiencies at the angles until 60 ° and thus less than 30 ° a constant shift can be observed in the data of both PMTs.

These are illustrated with red circles in Figure 5.20.

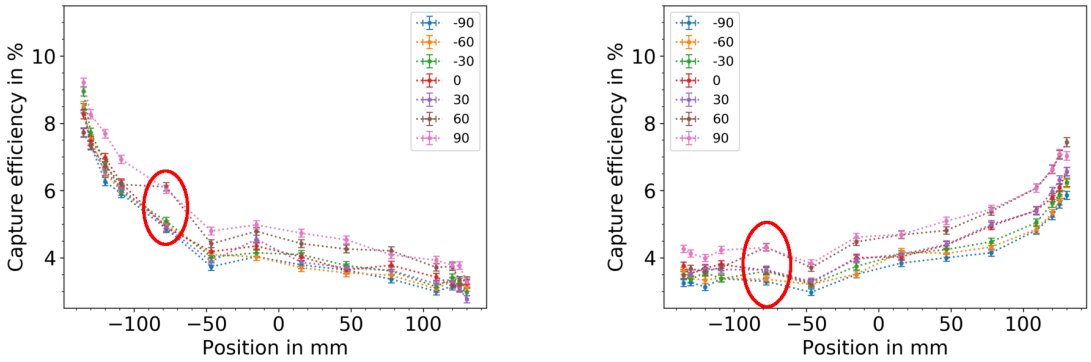


Fig. 5.20: One-sided capture efficiency of the WOM P45/3/5 measured with the left/right PMT on the left/right figure. The identified jump is illustrated with red circles.

To estimate its effect on the overall capture efficiency, $\langle\varepsilon\rangle_{z,\Phi}$ it was assumed that the increase of intensity at -80 mm is an effect of the light source only. The ratio of the one-sided capture efficiency of the angular averages $\langle\varepsilon\rangle_{\Phi\leq 30\,^\circ}(-80\text{ mm})$ and $\langle\varepsilon\rangle_{\Phi\geq 60\,^\circ}(-80\text{ mm})$ was estimated for both sides. The ratio was estimated as 1.185 ± 0.011 for the left 1.187 ± 0.014 for the right. The average of both is taken to correct all measured light intensities following from the time of the jump. The estimated overall efficiency changed from 8.25 % to 7.84 %. The new value is taken as the result P45/3/5 (see Table A.2). This relative change of the capture efficiency was about 5.2 %. With this method, only drastic jumps can be identified. There were observed only two (10 % of all recorded jumps) intensity jumps with an amplitude larger than 18 % (see Section 4.2.2). It can be expected that other WOM characterizations were compromised by this effect but were too small to be identified with the above method. Nevertheless, this deviation gives an upper limit for the uncertainty of this effect, which will be an overestimation. This error is quadratically added to the lower and upper systematic uncertainties discussed in Section 5.4.1. The jumps can be a property of the LED or an effect of the pulse generator. A variation in the line voltage is also possible.

5.5 Estimation of the overall capture efficiency

To estimate the overall efficiency, the measurements of both PMTs needed to be combined to get the two-sided efficiency. The WOM is assumed to be radial symmetric, and the angular average $\langle\mu_{eff}^{\text{PMT}}\rangle_\Phi(z)$ has been calculated for both PMTs. The resulting curves for the WOM P45/5-6/2 are shown in Figure 5.21. The angular average $\langle\mu_{eff}^{\text{PMT}}\rangle_\Phi(z)$ shown in orange (right PMT) and blue (left PMT) are the one-sided

efficiencies. The two-sided efficiency is the sum of both curves and is shown in green. The overall efficiency $\langle \mu_{eff} \rangle_{\Phi,z}$ is the average of all data point in the range between -100 mm and 100 mm of the WOM's radial axis and all measured angles. The data with a distance less than 50 mm to a PMT was excluded because the light distribution is not homogeneous (see Section 5.3) and the values are comprised by misdirected light (see Section 5.2.4).

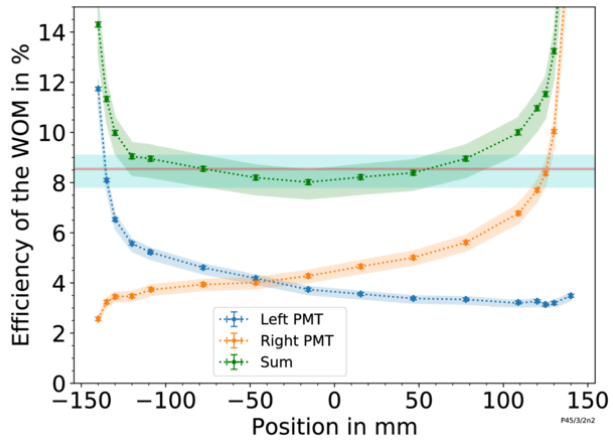


Fig. 5.21: The resulting two-sided (green) and the one-sided efficiencies (blue and orange) of the WOM P45/5-6/2 with the diameter of 45 mm and a wall thickness of 5.5 mm. The shaded region error is the systematic error discussed in 5.4.1. The red dashed line marks the averaged region.

The two-sided averaged capture efficiency $\langle \varepsilon \rangle_{\Phi,z}$ has been estimated for all investigated WOMs. The results are shown in Figure 5.22. A detailed list of all WOMs can be found in the appendix Table A.2. The systematic uncertainties given here are the quadratic sum of those caused by the inhomogeneous photocathodes of the PMTs estimated in Section 5.4.1 and the upper limit of the deviation caused by the shift from Section 5.4.3. The upper limit for the relative overestimation caused by the nonlinearity of the PMTs (see Section 5.4.2) lies between 12 % and 16 %. The maximum limit for the relative uncertainty caused by the jumps of the light source described in Section 4.2.2 is 5.2 %. If only the WOM tubes without special treatment are compared, no dependency on the wall thickness is observed. Furthermore, the WOMs with the wall thickness 4 mm but different diameters show no tendency at all. The quartz tubes do not differ from PMMA. It is observed that the rods are significantly better than the tubes. This behavior is consistent with the assumption that a larger surface generates more losses. A real surface has defects that increase the scattering losses. The easiest way to reduce these defects is to reduce the surface of the WOM. If only the tubes without special treatment are taken into account the average achieved efficiency is

$$(8.49 \ \pm 0.01_{\text{stat}} \ ^{+0.41}_{-0.55 \text{ sys}}) \ \%$$

This result is factor 8.6 smaller than the theoretical achievable efficiency of $\approx 73\,\%$ from Section 2.1.1. The overall efficiency of these WOMs varied with a standard

deviation of ±0.77 %. The PMMA tube coated twice has shown ≈ 3 % better capture efficiency than the average, and the quartz tube coated by hand has shown the best result with:

$$(18.51 \ \pm 0.07_{\text{stat}} \ ^{+1.39}_{-1.53 \text{ sys}}) \ \%$$

It can be concluded that the paint thickness was larger for the double coated WOM and those coated with higher speed. From this, it follows that the achieved paint thicknesses were not in the range at which the absorption coefficient a_{ABS} is 100 %. The assumption that the withdraw speed of 150 $\frac{\text{mm}}{\text{min}}$ at a temperature between 22 °C and 24 °C would be enough to achieve $a_{\text{ABS}} = 100\,\%$ was wrong. In [BK17] it was assumed that the three rods with a diameter of 20 mm were coated that a_{ABS} is 100 % as well. The rods have shown a favorable site (see Section 5.2.4). It is possible that this is caused by variations in the paint thickness. In this case, the assumption of complete absorption is wrong, and the paint thickness was still in the same order as its absorption length. Which could explain the low overall capture efficiencies.[9] Theoretically, the paints layer thickness increase proportional to the coating speed to the power of 1/2 or 2/3 depending on the viscosity [Bri13]. Besides the conditions such as temperature and current viscosity, it was expected that the absorption of the paint saturates to 100 % as long as the coating speed is high enough. In [Heb14] it was observed that with increasing coating speed the paint thickness saturates at a speed of 40 $\frac{\text{mm}}{\text{min}}$ to ≈ 20 μm (see Figure 3.2 in Section 3.2). Other dependencies were not investigated. Besides the unknown boundaries of the theory from [Bri13], the drying process is not considered. The WOMs were hanging during the drying as they were coated. At the beginning of the drying process, the paint is still fluid and has the tendency to stream downwards. If the paint thickness is larger, the gravitational force is larger and more paint flows down. This paint loss will depend on the viscosity and therefore on the temperature. It can be expected that there is a maximal achievable absorption that depends on external parameters such as temperature during coating and drying. The room temperature for the coating procedure used by Peiffer et al. in Mainz were 13 °C in average [Pei19]. In this work, the temperature during the coating was 10 °C higher. The actual toluene concentration has an effect on the viscosity too. By refilling the lost toluene (up to 9 % see Section 3.1) the viscosity of the paint was decreased. If the toluene loss was not considered in the work of [Heb14] and Peiffer et al., the actual viscosity was higher, which lead to higher capture efficiencies during higher absorption. It is possible that the lower viscosity in the coating process done in this work explains the systematical lower efficiencies of the WOMs compared with those produced in Mainz.

The PMMA tubes coated in the inner side as well showed the same behavior as

[9] Compare with usual achieved one-sided efficiencies of (24 ± 1) % at a distance of 150 mm to the PMT produced by Peiffer et al. [Bös18].

the other WOM tubes. Under the assumption that the absorption of the UV light was incomplete, the effective paint thickness of these tubes is larger, and a higher absorption can be expected if the PMMA and the matrix EMC of the paint are UV transparent. Thus it can be concluded that at least one of these materials is not transparent for the UV light around 375 nm.

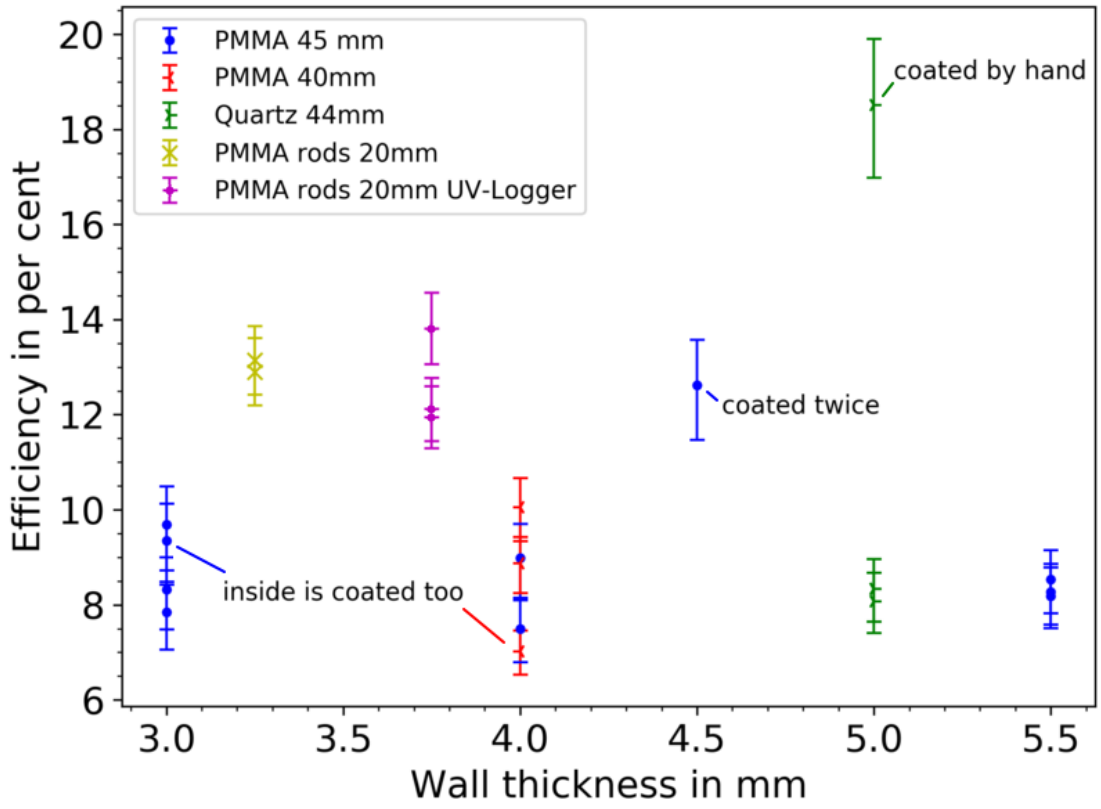


Fig. 5.22: Two-sided averaged capture efficiencies $\langle \varepsilon \rangle_{\Phi,z}$ of all investigated WOMs in dependency of the wall thickness. WOMs with additional treatment are marked. The error bars represent the systematic uncertainties only because the statistic uncertainties are negligible.

6. Cross check with different methods

To verify the results on the capture efficiency measurements, a cross-check has been done with a second setup at the University of Mainz. The capture efficiency of two WOMs coated in this work was characterized in this work and with the characterization setup in Mainz. The WOMs had a diameter of 45 mm and wall thicknesses of (3.0 ± 0.5) mm and (5.5 ± 0.5) mm.

6.1 Setup Mainz

Instead of counting photoelectrons, the setup estimates the current generated by the illuminated PMTs. To suppress the background, a Lock-in-Amplifier has been used. As a light source, a Xenon-Lamp has been used with a monochromator. The measurement was done at a wavelength of (375 ± 5) nm [RH19]. The used PMT is an IceCube-PMT[1] of the type R7081-02 from Hamamatsu. A two-dimensional surface scan was done on this PMT in this work (see Section 4.3). At the beginning of the measurement, the PMT is illuminated directly with the light source in its photocathode center for a reference value. In the next step, the WOM is illuminated at different positions on the radial axis. The angle of the light source was not varied. For further details about the setup, see [Pei18b, 5].

6.2 photocathode's spatial variation setup Mainz

To investigate the effect of the WOM alignment concerning the center of the photocathode in the setup in Mainz, the analysis done in Section 5.4.1 was also done on the data of the surface scan of the IceCube-PMT in Section 4.3. Next to the WOMs used for the cross-check with a diameter of 45 mm, a WOM with 90 mm in diameter was simulated too because this is the geometry of the prototype (see Section 2.1). The left of Figure 6.1 shows the relative sensitivity of the photocathode of the used PMT in the setup Mainz. Due to the resolution of the scan it was assumed that the alignment of 68 % of the WOMs is not better than 5 mm and that the wallthickness is 5 mm too. The right of Figure 6.1 shows the simulated distribution for the relative effective sensitive for both WOM diameters. The resulting expectation values for the relative sensitivities can be seen in Table 6.1. In the measurements done in Mainz for the cross-check, the photocathode center was used to estimate the reference

[1]Serialnumber: R7081-100-10

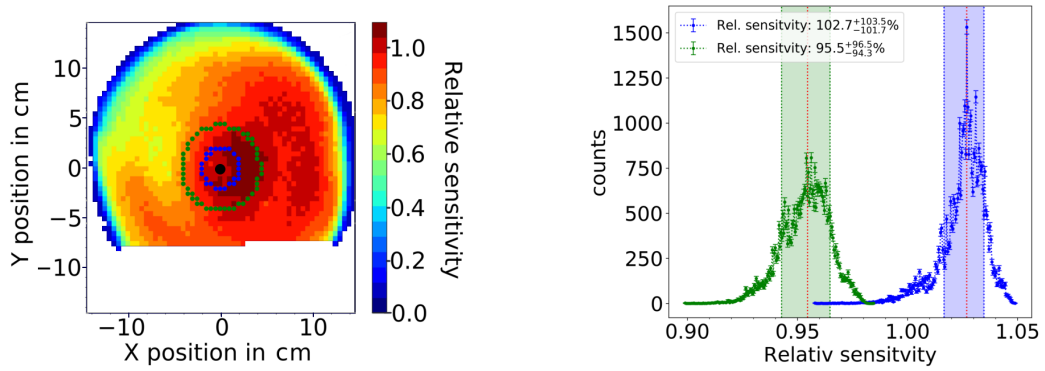


Fig. 6.1: Left: The two dimensional scan with the shape of a centered WOMs with the diameters of 45 mm (blue) and 90 mm (green). The center is marked with a black dot. Right: The distributions of the averaged relative sensitivity estimated in the montecarlo simulation for those WOMs.

Diameter in mm	Rel. sensitivity in %
45	$102.7^{+0.8}_{-1.0}$
90	$95.5^{+1.0}_{-1.2}$

Tab. 6.1: Results of the Monte-Carlo simulation for the WOMs with a diameter of 45 mm and 90 mm on the IceCube-PMT scan. The uncertainties are calculated with the quantiles Q_{16} and Q_{86} of the estimated distributions as it is described in Section 5.4.1. The uncertainty of the alignment is chosen to be 5 mm because of the finite resolution of the scan done in Section 4.3. The uncertainty estimated by [RH19] is 3 mm. The width of the distributions is therefore overestimated.

value. It was found that the relative sensitivity of those parts of the photocathode covered by a WOM with 45 mm is $2.7^{+0.8}_{-1.0}$ % more efficient than the center. As a result, the estimated capture efficiencies are overestimated. In the case of a 90 mm WOM, the covered area is $4.5^{+1.0}_{-1.2}$ % less efficient than the center, which would lead to an underestimation. For the reference value, the light source has a distance of approximately 25 mm to the photocathode. The light cone's diameter is assumed to be 20 mm [RH19]. The covered area of this circle has a relative sensitivity of $(102.7 \pm 0.2_{\text{stat}})$ %. Because the light's real distribution is not estimated, it is not known how the relative sensitivity has to be weighted to calculate the actual response of the PMT. Nevertheless, this value is an upper boundary for the response. A radial symmetric light distribution with a larger diameter would have more overlap with the more sensitive region given by the alignment of the first dynode [Abb10]. The relative response is, therefore, monotonic, increasing with the increasing width of the light cone. If the center $(100.0 \pm 0.7_{\text{stat}})$ % is the lower limit of the response for light cone with a diameter ≤ 5 mm, the overestimation of the 45 mm WOM lies therefore

between:

$$0^{+0.9}_{-1.0}\ \% \text{ and } 2.7^{+1.1}_{-1.2}\ \%$$

In the case of 90 mm the underestimation is between:

$$4.5^{+1.2}_{-1.4}\ \% \text{ and } 7.2^{+1.0}_{-1.2}\ \%$$

A proper correction requires the light distribution to be measured. Alternatively, the weighted average of both ranges could be taken with the boundaries and its uncertainties as the uncertainty for the average.
In the case of the cross-check, the overestimation of the efficiency of the 45 mm WOMs is too small to have a significant effect. In the case of the 90 mm WOMs, the inhomogeneity must be taken into account if the relative uncertainty shall be smaller than 7.2 %.

6.3 Results

For the cross-check, two PMMA tubes were used with a wall thickness of 3 mm and 5 mm and a diameter of 45 mm. Both tubes were coated as described in Chapter 3 and were characterized in the setup Zeuthen first and measured later in Mainz. Figure 6.2 shows the results of the one-sided capture efficiencies $\varepsilon_{\text{eff}}^{\text{PMT}}(z)$ as a function of the distance to the detecting PMT. The efficiency measured in Zeuthen is an average $\langle\varepsilon_{\text{eff}}^{\text{PMT}}\rangle_{\Phi}(z)$ over the different angles of illumination Φ and plotted for both PMTs separately. The results from Mainz are the one-sided capture efficiency measured for only one angle of illumination from one PMT. The measurement in Mainz was done without information about the orientation of the WOM. It is therefore not clear which one-sided capture efficiency must be compared with those from Mainz. To compare the measurements, the average of all one-sided capture efficiencies has been calculated in the range from −100 mm and 100 mm. The averages are listed in Table 6.2.

wall thickness:	(3.0 ± 0.5) mm	(5.5 ± 0.5) mm
Mainz:	3.14 ± 0.15	3.71 ± 0.25
Zeuthen left:	$3.69^{+0.12}_{-0.21}$ (2.1σ)	$3.90^{+0.09}_{-0.15}$ (0.7σ)
Zeuthen right:	$4.48^{+0.24}_{-0.28}$ (4.2σ)	$4.59^{+0.19}_{-0.23}$ (2.6σ)

Tab. 6.2: Averages of the one-sided capture efficiencies in the range between −100 mm and 100 mm of the radial axis of the measurements done in the crosscheck with the WOMs P45/3/2 and P45/5-6/1. The values in braces are the difference of the averages of the one-sided efficiency measured in Zeuthen and Mainz in multiples of σ.

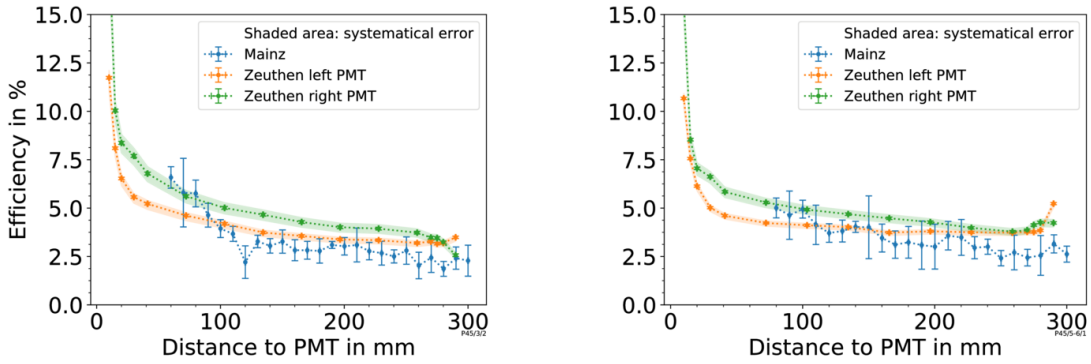


Fig. 6.2: Crosscheck with Mainz. Both WOMs have a diameter of 45 mm Left: WOM P45/3/2 with a wall thickness of (3.0 ± 0.5) mm. Right: WOM P45/5-6/1 with a wall thickness of (5.5 ± 0.5) mm.

The averages of the WOM with 5 mm are in agreement with both PMTs.The left one-sided efficiency of the WOM with 3 mm wall thickness measured in Zeuthen is in agreement with those done in Mainz. The deviation between the right PMT in Zeuthen and the measurement from Mainz is 4.2σ. By taking the nonlinearity of the PMTs used in Zeuthen investigated in Section 5.4.2 into account, the discrepancy is reduced. The possible overestimation lies between 12 % (left PMT) and 16 % (right PMT), which can partially explain the discrepancy. Even if the proper side to account is not known, the two measurements (Zeuthen and Mainz) are in good agreement. This confirms the results of Section 5.5 with an absolute uncertainty of $\leq 2\,\%$ if the discrepancy of the two-sided capture efficiency of the 3 mm WOM is used to estimate this value.

7. Summary

A setup to estimate the capture efficiency of an optical module of arbitrary curved shape was developed. For this, a software package was written to move a light source with the Five Axis Scanner pointing at arbitrary points of the modules' surface. To automatize the measurement, the software finds the optimal trajectory by itself while avoiding collisions (see Chapter 4).

Twenty-five WOMs of different geometries and materials (PMMA, quartz) were coated in this work. The coating process (see Chapter 3) was standardized to show the capture efficiency's reproducibility. To prevent the WOM tubes from being filled and coated inside, they were closed at both ends with aluminum caps sealed with PTFE. It was observed that the required sealing force is strong enough to compress the PMMA tubes by $\frac{1}{300}$ of their length. The use of expanded PTFE avoided those contractions (see Section 3.5). By estimating the paint thickness of the PMMA rods, it was discovered that more PMMA had been solved in the paint than paint was coated on the tube, and an estimation of the paint thickness by weighing with PMMA as a substrate is not possible (see Section 3.4). A cross-check with the setup at the University of Mainz was done with two WOMs. The results of both measurements were in agreement within an absolute uncertainty of $\leq 2\,\%$ in the two-sided capture efficiency (see Chapter 6). Twenty-two WOMs were characterized. The averaged deviation from the angular average σ_Φ of WOMs was $4.2\,\%$ with a fluctuation of $1.6\,\%$ (see Section 5.2.4). The light distribution on both ends of one WOM tube was measured, and it was concluded that it can be seen as homogeneous over the whole end for illuminations with distances $\geq (50 \pm 10)$ mm to the PMTs.

The average of the estimated overall two-sided capture efficiency $\langle \varepsilon_{eff} \rangle_{\Phi,z}$ which is the product of the transport efficiency ε_T, the geometrical capture efficiency ε_CPT given by the angle of total reflection, the conversion efficiency of the paint ε_WLS and the absorption coefficient of the paint a_WLS for the investigated WOM tubes was $(8.49 \ \pm 0.01_\mathrm{stat}\ ^{+0.41}_{-0.55\,\mathrm{sys}})\ \%$. The two-sided efficiencies of those WOMs varied with a standard deviation of $\pm 0.77\,\%$. The rods coated in this work were significantly better with $(13.0 \ \pm 0.1_\mathrm{stat}\ ^{+0.7}_{-0.7\,\mathrm{sys}})\,\%$ on average. No dependencies on the wall thickness or the tube's diameter were observed (see Section 5.5).

Coating the tube twice had increased the efficiency and coating by hand had given the best result with $(18.51 \ \pm 0.07_\mathrm{stat}\ ^{+1.39}_{-1.53\,\mathrm{sys}})\ \%$. From this, it was concluded in Section 5.5 that the achieved paint thickness was not in the range at which the UV light is absorbed completely by the paint ($a_\mathrm{WLS} < 100\,\%$). The viscosity of the paint during the coating and drying process might be the most promising

candidate to explain the discrepancy with the two-sided efficiencies up to $(48 \pm 2)\,\%$ for WOMs produced and characterized from Böser et al. A low viscosity due to higher environmental temperature and/or higher solvent (toluene) concentration in the paint leads to less paint thickness during the coating. While the paint is drying, less viscosity has a higher tendency to flow downwards. To gain comparability, the actual viscosity of the paint under the conditions of the coating and drying process must be determined. The additional coating at the inner side of two PMMA tubes did not show any observable effect. It was concluded that the PMMA and/or the paint's matrix (EMC) is not UV transparent due to impurities. To exclude that the PMMA also absorbs parts of the paint's short wavelength emission spectrum, the absorption length for different wavelengths must be determined with material from the same manufacturer. Additionally, a chemical analysis of the impurities of the material is recommended.

The actual paint thickness of a WOM and the absorption length of the paint is still unknown. The determination of the achieved paint thickness could be estimated by weighing light guides made out of quartz. In the future, the absorption length must be determined to calculate the minimal paint thickness.

This work has shown that the light guides of the WOMs can be reproducibly coated with an absolute variation in the overall two-sided capture efficiency of $0.8\,\%$ and that the variation of the diameter in the range of $d \in [40\text{ mm}, 45\text{ mm}]$ and of the wall thickness $w \in [3\text{ mm}, 5.5\text{ mm}]$ have no significant influence on the efficiency. Compared with the tube geometry, the rods show $\geq 50\,\%$ higher efficiencies.

Acknowledgement

In the beginning I want to thank Professor Marek Kowalski who gave me the opportunity to join the IceCube group. I also want to acknowledge him and Timo Karg for their support and constructive advices at each time. Furthermore I want to thank Dustin Hebecker and Jannes Brostean-Kaiser that they have shared their WOM experience with me during the whole work and Daniel Küsters for his support in the end. I am very glad that Peter Peiffer has introduced me in the common coating process and Sebastian Böser, John Rack-Helleis and Anna Steuer from the group of experimental particle and astroparticle physics for answering questions and constructive discussions. More thanks goes to Lap Van Vu who has supported me every time I needed his engineering skills and Ronny Sternberger who build up the control unit of the portal robot and has written its API. Special thanks to Timo, Daniel, Jannes, Dustin, Summer and Jakob for reviewing. In the end I want to thank all other group members of the IceCube group in DESY Zeuthen which have created such a constructive and pleasant atmosphere.

A. Appendix

Diameter d in mm	Wall thickness w in mm	Rel. sensitivity [%]	Rel. error [%]
5.0	−*	99.2	$-1.2/+1.3$
10.0	−*	98.7	$-0.7/+0.7$
20.0	−*	97.1	$-0.3/+0.2$
40.0	4.0	91.2	$-2.1/+1.1$
44.0	5.0	84.4	$-3.7/+2.4$
45.0	3.0	75.7	$-5.6/+3.2$
45.0	4.0	78.9	$-4.8/+2.8$
45.0	4.5	80.2	$-4.6/+2.7$
45.0	5.5	82.2	$-3.8/+2.3$

(a) Results for the left PMT.

Diameter d in mm	Wall thickness w in mm	Rel. sensitivity [%]	Rel. error [%]
5.0	−*	99.1	$-1.4/+1.0$
10.0	−*	97.4	$-0.6/+0.4$
20.0	−*	94.9	$-0.2/+0.2$
40.0	4.0	87.3	$-3.9/+2.6$
44.0	5.0	79.8	$-5.0/+4.6$
45.0	3.0	70.7	$-6.2/+5.3$
45.0	4.0	74.0	$-6.0/+5.1$
45.0	4.5	75.6	$-5.7/+4.5$
45.0	5.5	77.6	$-5.0/+4.1$

(b) Results for the right PMT.

Tab. A.1: Results of the Monte Carlo simulation to find the relative error due to the inhomogeneity of the photocathode of the left PMT (*: rods). The relative sensitivity is the averaged sensitivity of the area of the photocathode covered by the WOM with respect to its center.

A.1 Results of all investigated WOMs

Name	d in mm	w in mm	Overall two-sided capture efficiency $\langle\epsilon\rangle_{z,\Phi}$ in %	Two-sided capture efficiency in the center $\langle\epsilon\rangle_{\Phi}(z=0)$ in %
P20/1[(1)]	20.0	−*	$12.11 \pm 0.02_{\text{stat}}\ ^{+0.66}_{-0.66\ \text{sys}}$	$11.47 \pm 0.08_{\text{stat}}\ ^{+0.60}_{-0.60\ \text{sys}}$
P20/2[(2)]	20.0	−*	$11.95 \pm 0.02_{\text{stat}}\ ^{+0.65}_{-0.65\ \text{sys}}$	$10.99 \pm 0.08_{\text{stat}}\ ^{+0.57}_{-0.57\ \text{sys}}$
P20/3[(3)]	20.0	−*	$13.81 \pm 0.02_{\text{stat}}\ ^{+0.75}_{-0.76\ \text{sys}}$	$13.03 \pm 0.08_{\text{stat}}\ ^{+0.68}_{-0.68\ \text{sys}}$
P20/4	20.0	−*	$12.90 \pm 0.10_{\text{stat}}\ ^{+0.70}_{-0.71\ \text{sys}}$	$12.87 \pm 0.19_{\text{stat}}\ ^{+0.67}_{-0.67\ \text{sys}}$
P20/5	20.0	−*	$13.14 \pm 0.10_{\text{stat}}\ ^{+0.72}_{-0.72\ \text{sys}}$	$13.15 \pm 0.19_{\text{stat}}\ ^{+0.69}_{-0.69\ \text{sys}}$
P40/4/2	40.0	4.0	$10.05 \pm 0.04_{\text{stat}}\ ^{+0.62}_{-0.70\ \text{sys}}$	$10.31 \pm 0.10_{\text{stat}}\ ^{+0.61}_{-0.70\ \text{sys}}$
P40/4/3	40.0	4.0	$8.87 \pm 0.10_{\text{stat}}\ ^{+0.54}_{-0.62\ \text{sys}}$	$8.82 \pm 0.24_{\text{stat}}\ ^{+0.52}_{-0.60\ \text{sys}}$
P40/4/5[(4)]	40.0	4.0	$7.02 \pm 0.04_{\text{stat}}\ ^{+0.43}_{-0.49\ \text{sys}}$	$6.63 \pm 0.08_{\text{stat}}\ ^{+0.39}_{-0.45\ \text{sys}}$
P45/3/2	45.0	3.0	$8.32 \pm 0.04_{\text{stat}}\ ^{+0.68}_{-0.83\ \text{sys}}$	$8.22 \pm 0.10_{\text{stat}}\ ^{+0.66}_{-0.81\ \text{sys}}$
P45/3/3	45.0	3.0	$9.69 \pm 0.05_{\text{stat}}\ ^{+0.80}_{-0.97\ \text{sys}}$	$9.49 \pm 0.12_{\text{stat}}\ ^{+0.43}_{-0.59\ \text{sys}}$
P45/3/4[(5)]	45.0	3.0	$9.35 \pm 0.05_{\text{stat}}\ ^{+0.77}_{-0.93\ \text{sys}}$	$9.28 \pm 0.10_{\text{stat}}\ ^{+0.75}_{-0.91\ \text{sys}}$
P45/3/5	45.0	3.0	$7.84 \pm 0.03_{\text{stat}}\ ^{+0.65}_{-0.78\ \text{sys}}$	$7.91 \pm 0.06_{\text{stat}}\ ^{+0.64}_{-0.78\ \text{sys}}$
P45/4-5/1[(6)]	45.0	4.5	$12.62 \pm 0.03_{\text{stat}}\ ^{+0.96}_{-1.15\ \text{sys}}$	$12.44 \pm 0.06_{\text{stat}}\ ^{+0.92}_{-1.12\ \text{sys}}$
P45/4/1	45.0	4.0	$8.99 \pm 0.02_{\text{stat}}\ ^{+0.71}_{-0.85\ \text{sys}}$	$9.08 \pm 0.05_{\text{stat}}\ ^{+0.71}_{-0.84\ \text{sys}}$
P45/4/2	45.0	4.0	$7.50 \pm 0.05_{\text{stat}}\ ^{+0.60}_{-0.71\ \text{sys}}$	$7.62 \pm 0.11_{\text{stat}}\ ^{+0.59}_{-0.71\ \text{sys}}$
P45/5-6/1	45.0	5.5	$8.53 \pm 0.04_{\text{stat}}\ ^{+0.62}_{-0.71\ \text{sys}}$	$8.42 \pm 0.08_{\text{stat}}\ ^{+0.59}_{-0.69\ \text{sys}}$
P45/5-6/2	45.0	5.5	$8.19 \pm 0.03_{\text{stat}}\ ^{+0.59}_{-0.68\ \text{sys}}$	$8.14 \pm 0.06_{\text{stat}}\ ^{+0.57}_{-0.66\ \text{sys}}$
P45/5-6/3	45.0	5.5	$8.27 \pm 0.02_{\text{stat}}\ ^{+0.60}_{-0.69\ \text{sys}}$	$8.27 \pm 0.06_{\text{stat}}\ ^{+0.58}_{-0.68\ \text{sys}}$
Q44/5/2	44.0	5.0	$8.33 \pm 0.04_{\text{stat}}\ ^{+0.63}_{-0.69\ \text{sys}}$	$8.37 \pm 0.08_{\text{stat}}\ ^{+0.61}_{-0.68\ \text{sys}}$
Q44/5/4	44.0	5.0	$8.33 \pm 0.05_{\text{stat}}\ ^{+0.63}_{-0.69\ \text{sys}}$	$8.32 \pm 0.09_{\text{stat}}\ ^{+0.61}_{-0.67\ \text{sys}}$
Q44/5/3	44.0	5.0	$8.07 \pm 0.05_{\text{stat}}\ ^{+0.61}_{-0.67\ \text{sys}}$	$8.03 \pm 0.09_{\text{stat}}\ ^{+0.59}_{-0.65\ \text{sys}}$
Q44/5/5[(7)]	44.0	5.0	$18.51 \pm 0.07_{\text{stat}}\ ^{+1.39}_{-1.53\ \text{sys}}$	$18.73 \pm 0.14_{\text{stat}}\ ^{+1.37}_{-1.52\ \text{sys}}$

Tab. A.2: With d as the outer diameter of the WOM and w as the wall thickness (*: rods). The overall two-sided capture efficiency $\langle\epsilon\rangle_{z,\Phi}$ is an average over all angles Φ and $z \in [-100\ \text{mm}, 100\ \text{mm}]$. The efficiency $\langle\epsilon_{z=0}\rangle_{\Phi}$ is the average over all angles Φ at the center of the WOM which has a distance of 150 mm to each PMT. In the case of the WOMs P20/1, P20/2 and P20/3 the distance is 250 mm. The systematic uncertainties are the quadratic sum of the variations due to the alignment of the inhomogeneous photocathode from Section 5.4.1 listed in Table A.1 and the uncertainty from the light source discussed in Section 4.2.2. Due to the nonlinearity of the PMTs (see Section 5.4.2) the efficiencies might be overestimated by a relative factor between 12 % and 16 %. (1), (2) and (3): Coated in the work of [BK17]. (4) and (5): Coated at the inner side as well. (6): WOM is coated twice. (7): Hand coated WOM.

A.2 Instructions to use WACKER SilGel® 612 A/B for optical contact

For optical contact between the PMTs and the WOMs SilGel® 612 A/B[1] was used. It is a vulcanizable two component silicone rubber. It is a transparent gel with an optical density of 1.404 [AG99]. There are two possible options to use the Gel for optical contact. The first way is to vulcanize the components mixture directly on the window in front of the photocathode formed by a reflecting casting mould. The second is to produce a pad of the wanted size and thickness and bring it in contact with the PMTs window after vulcanization. The first method has the advantage that the contact will be as good as possible. Air between the cathode and the can be avoided easily. In contrast if the Gel is damaged or contaminated with dust repeating the procedure is sophisticated. With the second methode one is able to produce multiple pads which can be exchanged when needed. Light can be lost at the cutting edge of the pad because it has no optical contact to a reflecting area.
Component A are the monomers and component B is the catalyser. The manufacturer recommend a ratio of the components of 1 : 1. Nevertheless in this work a ratio of three parts of A to two parts of B has been used to have less elasticity. This ratio is an approved ratio used in the IceCube-DOMs and the measurement setup in Mainz [Pei18a]. The other advantage that one gains more procssing time for filling the mixture into its mould. After mixing the components together the maximal processing time is one hour. After this time the viscosity becomes to high to fill it into a form. Full vulcanization has ended after eight hours. It is recommended to wheigh out the needed amount for both components with a scale before mixing in separate beakers instead of using measuring cylinders. It is important to note that the component A should never be contaminated with traces of component B if it shall not be vulcanized. Remember which tool had contact with B and use new clean tools for A if in doubt. Before mixing the mould has to have been prepared. When mixing the components careful stiring with a spoon spatula is recommended. Producing bubbles should be avoided by holding the beaker of the first component slightly sloped and flowing the second component over the inner wall. The string takes 5 to 10 min. Bubbles can be removed by using a vibrating table of ≈ 10 min or by using a vacuum bell.
To vulcanize the gel at the window of the PMT a casting mould is needed. The mould should be reflective to avoid an additional photon loss. In this work the PMT XP2020 has a cylinder shaped form and a plane window in front of the photo cathode. Black tape was used to build an edge which worked as mould. The upper part of the tape was covered with aluminum foil (see figure A.1).

[1] Manufacturer Wacker Chemie AG

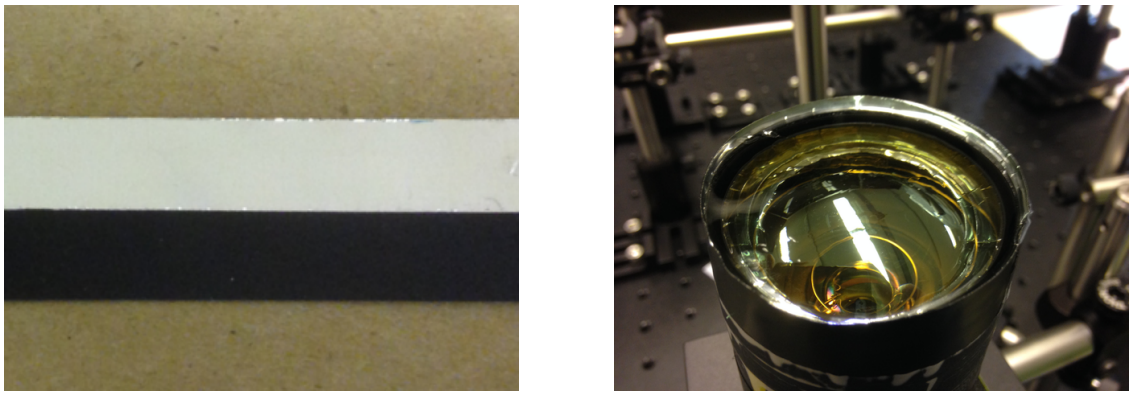


Fig. A.1: Left: Picture of black tape wth aluminum foil prepared to work as casting mould. Right: Prepared PMT with vulcanized SilGel® 612.

To produce gel pads a reversed culture dishe is masked by tape (see left of Figure A.2) and filles the mixture in. The dishes and the tape needs to be non-lipid. If needed a vibrating table can be used to remove bubbles and to distribute the gel homogeneously. After vulcanization has ended the tape can be removed. The pad itself can be removed from the dishes carefully by hand as well (see right of Figure A.2).

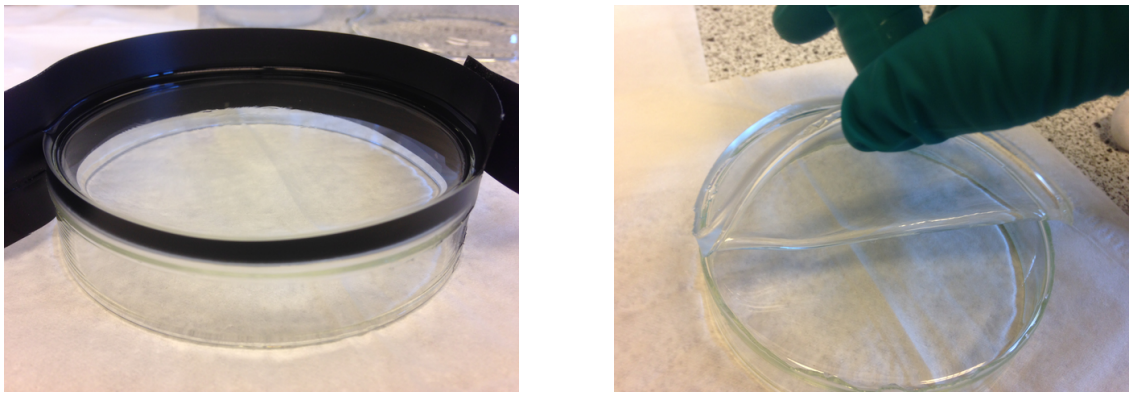


Fig. A.2: Left: Picture of black tape wth aluminium foil prepared to work as casting mould. Right: Prepared PMT with vulcanized SilGel® 612.

After filling the mixture into its mould the gel has to be covered to save it from dust. Directly after emptying the beaker with the vulcanizing gel the beaker must be cleaned with paper towel first without and later on with acetone or isopropanol.

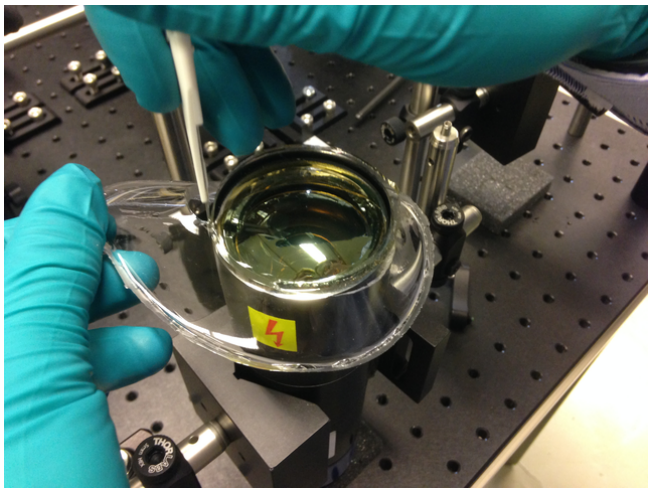


Fig. A.3: Produced gel pad out of SilGel® 612 cutted in size with scalpel.

A.3 Derivation of the torque from tube contraction

To derive the used torque on the screws one first calculates the contracting force F_d. The stiffness of acrylic plastic is $3.2 \times 10^3 \frac{\mathrm{N}}{\mathrm{mm}^2}$ [Kuc96]. F_d follows from Hooks law [Kuc96]:

$$F_d = E\frac{\Delta l}{l}A \tag{A.1}$$

with $\Delta l/l$ as the relative contraction of $1/300$ and A the cross section of the tubes which varied from 350 mm^2 to 630 mm^2 depending on the diameter and the wall thickness. It follows that the down warded force per nut F_d had to be between 1.3 kN and 2.2 kN. The acting force in the plain of the rotating nut F_r is perpendicular to F_d. Figure A.4 shows the decomposition of F_r. First it results in a force parallel to the unrolled thread $\cos(\alpha) \cdot F_r$ which can further decomposed to the downwarded force $F_d = \sin(\alpha)\cos(\alpha) \cdot F_r$. The torque M can calculated with:

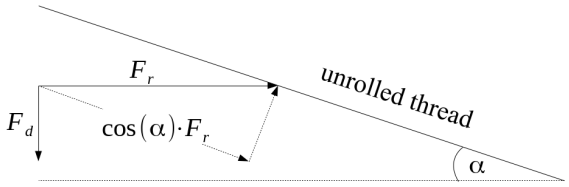


Fig. A.4: The decomposition of the force F_r acting on the thread with a pitch angle of α. The resulting downwarded force is F_d

$$M = \frac{2F_d}{\sin(2\alpha)} \cdot r \tag{A.2}$$

It follows with the radius of $r = 2.5$ mm of the M5 thread and a pitch angle α of $17\,^\circ$ the that the needed torque was between 11 Nm and 20 Nm for each nut.

A.4 Technical drawings

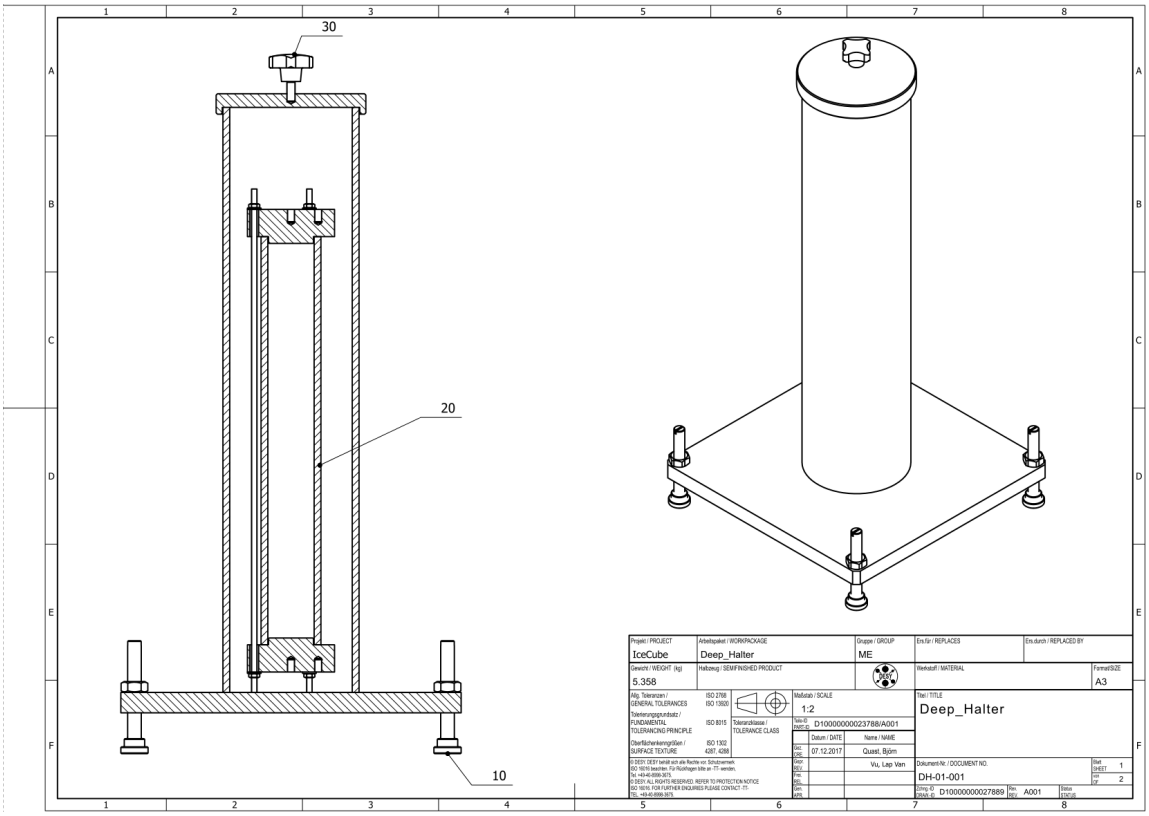


Fig. A.5: Technical drawing of the paint vessel. The vessel has a height of 430 mm, wall thickness of 5 mm and an inner diameter of 90 mm. The drawing is drawn by Lap Van Vu.

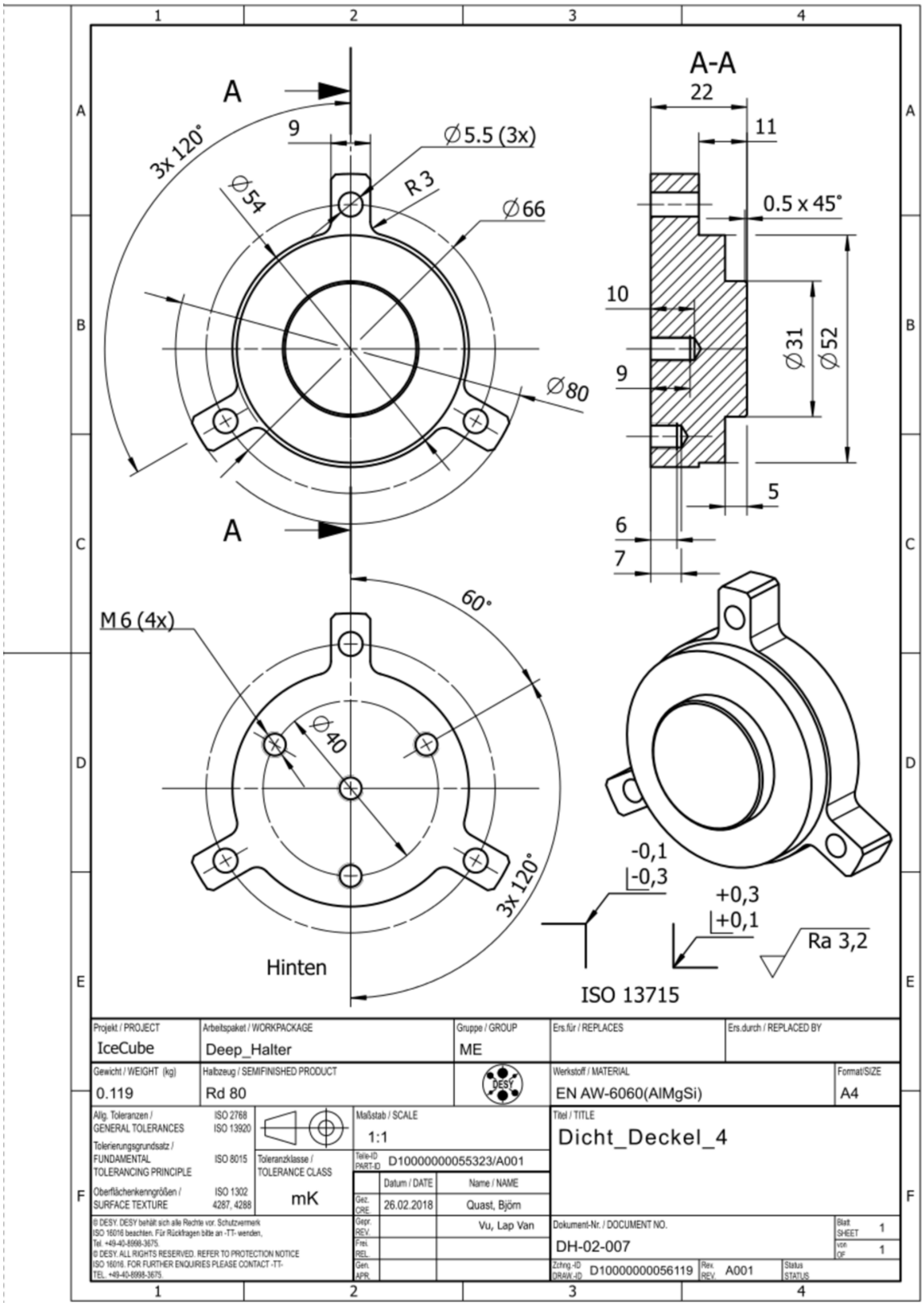


Fig. A.6: Technical drawing of the upper cap to close the light guiding tubes. The drawing is drawn by Lap Van Vu.

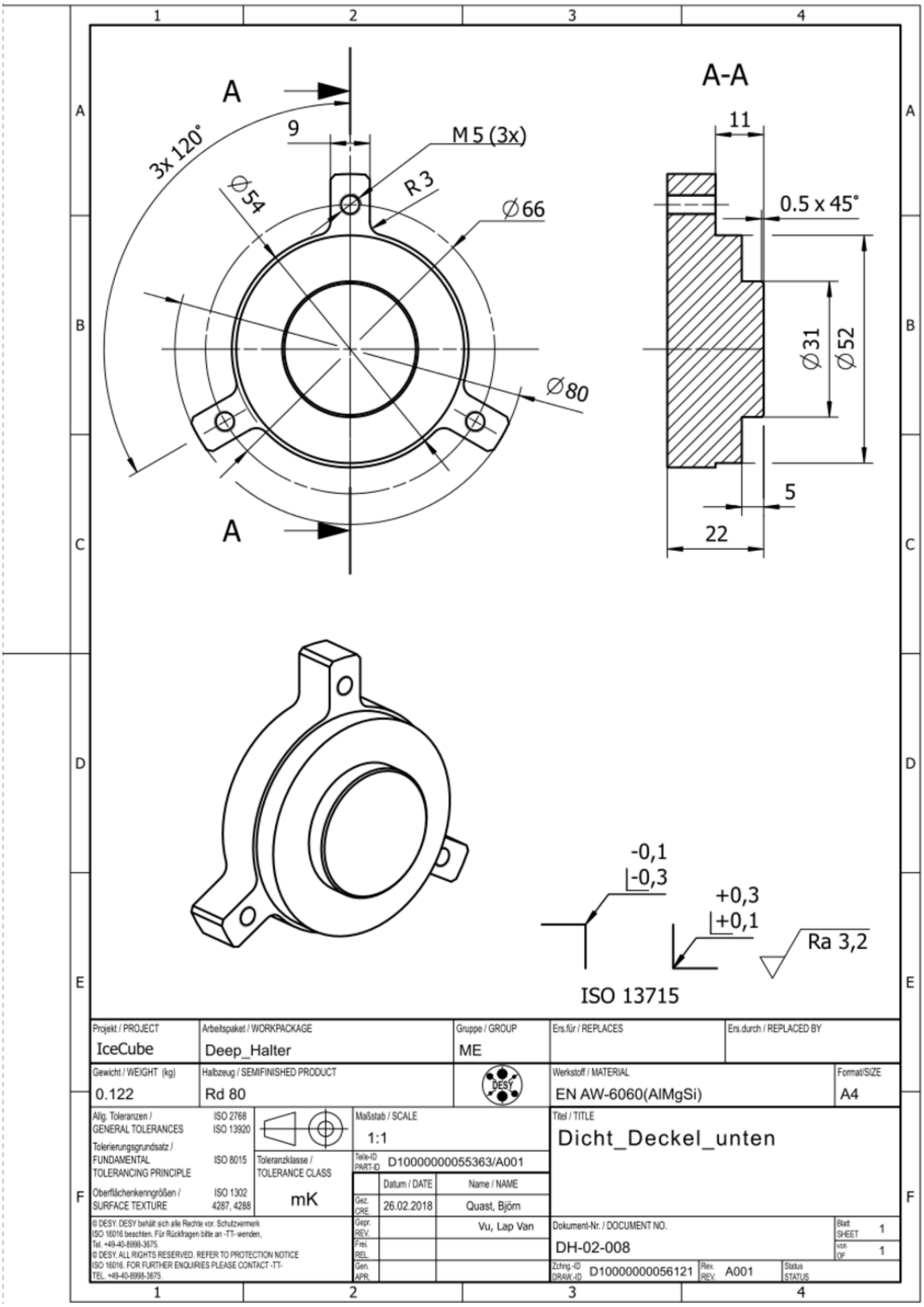


Fig. A.7: Technical drawing of the lower cap to close the light guiding tubes. The drawing is drawn by Lap Van Vu.

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Erklärung

Ich versichere, dass ich diese Arbeit selbständig verfasst und keine anderen als die angegeben Quellen und Hilfsmittel benutzt habe.

Berlin, October 18, 2026 .....

(Benjamin Bastian)

Declaration

I hereby declare that this thesis was formulated by myself and that no sources or tools other than those cited were used.

Berlin, October 18, 2026 .....

(Benjamin Bastian)